

**Fabrication and Performance Analysis of 3D-Printed Double Arrowhead Auxetic  
Structure for Impact Mitigation**

Submitted By

**Maysha Anan Aishy (200012133)**

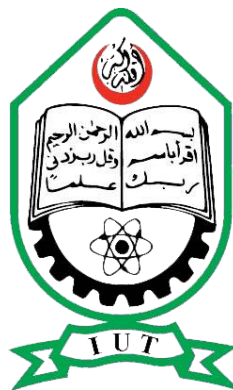
**Naoshin Anzum Islam (200012146)**

**Zahin Tasnim Ridita (200012153)**

Supervised By

**Prof. Dr. Md. Zahid Hossain**

**A Thesis submitted in partial fulfillment of the requirement for the degree of Bachelor  
of Science in Mechanical Engineering**



**Department of Mechanical and Production Engineering (MPE)**

**Islamic University of Technology (IUT)**

**September, 2025**

### Candidate's Declaration

This is to certify that the work presented in this thesis, titled "Fabrication and Performance Analysis of 3D-Printed Double Arrowhead Auxetic Structure for Impact Mitigation," is the outcome of the investigation and research carried out by me under the supervision of Prof. Dr. Md. Zahid Hossain, Professor, Department of Mechanical and Production Engineering, Islamic University of Technology.

It is also declared that neither this thesis nor any part of it has been submitted elsewhere for the award of any degree or diploma.

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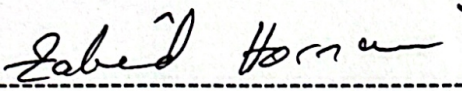
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# Recommendation of the Thesis Supervisors

The thesis titled "Fabrication and Performance Analysis of 3D-Printed Double Arrowhead Auxetic Structure for Impact Mitigation" submitted by Maysha Anan, Student No: 200012133, Naoshin Anzum Islam, Student No: 200012146 and Zahin Tasnim Ridita, Student No: 200012153 has been accepted as satisfactory in partial fulfillment of the requirements for the degree of B Sc. in Industrial and Production Engineering on 29th September, 2025.

1. 

**Dr. Md. Zahid Hossain**

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# CO-PO Mapping of IPE 4800 -Thesis and Project

| COs | Course Outcomes (CO) Statement                                                                                                                                                                                                                                                                       | (PO)                                   | Addressed by                |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------|---|
| CO1 | <u>Discover and Locate</u> research problems and illustrate them via figures/tables or projections/ideas through field visit and literature review and <u>determine/Setting</u> aim and objectives of the project/work/research in specific, measurable, achievable, realistic and timeframe manner. | PO2<br>Problem analysis                | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     |   |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO2 | <u>Design</u> research solutions of the problems towards achieving the objectives and its application. Design systems, components or processes that meets related needs in the field of mechanical engineering                                                                                       | PO3<br>Design/development of solutions | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO3 | <u>Review, debate, compare</u> and <u>contrast</u> the relevant literature contents. Relevance of this research/study. Methods, tools, and techniques used by past researchers and justification of use of them in this work.                                                                        | PO4<br>Investigation                   | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO4 | <u>Analyse</u> data and <u>exhibit</u> results using tables, diagrams, graphs with their interpretation. <u>Investigate</u> the designed solutions to solve the problems through case study/survey study/experimentation/simulation using modern tools and techniques.                               | PO5<br>Modern tool usage               | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     |   |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO5 | <u>Apply</u> moral values and research/professional ethics throughout the work, and <u>justify</u> genuine referencing on sources, and demonstration of own contribution.                                                                                                                            | PO8<br>Ethics                          | Thesis Book                 |   |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill |   |
| CO6 | <u>Perform</u> own self and <u>manage</u> group activities from the beginning to the end of the research/work as a quality work.                                                                                                                                                                     | PO9<br>Individual work and teamwork    | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO7 | <u>Compile and arrange</u> the work outputs, write the report/thesis, a sample journal paper, and present the work to wider audience using modern communication tools and techniques.                                                                                                                | PO10<br>Communication                  | Thesis Book                 | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     |   |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill | ✓ |
| CO8 | <u>Recognize</u> the necessity of life-long learning in career development in dynamic real-world situations from the experience of completing this project.                                                                                                                                          | PO12<br>Life-long learning             | Thesis Book                 |   |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Performance by research     | ✓ |
|     |                                                                                                                                                                                                                                                                                                      |                                        | Presentation and soft skill |   |

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# **K-P-A Mapping of IPE 4800 -Thesis and Project**

| C<br>O<br>s | P<br>O<br>s  | Related Ks |        |        |        |        |        |        |        | Related Ps |        |        |        |        |        |        | Related As |        |        |        |        |
|-------------|--------------|------------|--------|--------|--------|--------|--------|--------|--------|------------|--------|--------|--------|--------|--------|--------|------------|--------|--------|--------|--------|
|             |              | K<br>1     | K<br>2 | K<br>3 | K<br>4 | K<br>5 | K<br>6 | K<br>7 | K<br>8 | P<br>1     | P<br>2 | P<br>3 | P<br>4 | P<br>5 | P<br>6 | P<br>7 | A<br>1     | A<br>2 | A<br>3 | A<br>4 | A<br>5 |
| C<br>O<br>1 | P<br>O<br>2  | √          | √      | √      | √      |        |        |        |        | √          |        |        |        |        |        |        |            |        |        |        |        |
| C<br>O<br>2 | P<br>O<br>3  |            |        |        |        | √      |        |        |        | √          | √      | √      |        |        |        | √      | √          | √      | √      |        |        |
| C<br>O<br>3 | P<br>O<br>4  |            |        |        |        |        |        |        | √      | √          | √      | √      |        |        |        |        | √          | √      |        |        |        |
| C<br>O<br>4 | P<br>O<br>5  |            |        |        |        |        | √      |        |        | √          |        |        |        | √      |        |        |            |        |        |        |        |
| C<br>O<br>5 | P<br>O<br>8  |            |        |        |        |        |        | √      |        |            |        |        |        |        |        |        |            |        |        |        |        |
| C<br>O<br>6 | P<br>O<br>9  |            |        |        |        |        |        |        |        |            |        |        |        |        |        |        |            |        |        |        |        |
| C<br>O<br>7 | P<br>O<br>10 |            |        |        |        |        |        |        |        |            |        |        |        |        |        |        |            |        |        |        |        |
| C<br>O<br>8 | P<br>O<br>11 |            |        |        |        |        |        |        |        |            |        |        |        |        |        |        |            |        |        |        |        |

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## **List of Sustainable Development Goals (SDGs) Addressed in this Project**

| SDG No. | Goals                      | Targets                                                                                                             | Relevance to the Thesis (put √ if valid) | Remarks            |
|---------|----------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------|
| 1       | No Poverty                 | 1.1 Eradicate extreme poverty (people living on less than \$1.25/day).                                              |                                          |                    |
|         |                            | 1.2 Reduce poverty in all its forms by at least half.                                                               |                                          |                    |
|         |                            | 1.3 Implement nationally appropriate social protection systems.                                                     |                                          |                    |
|         |                            | 1.4 Ensure equal rights to economic resources, services, property, inheritance, technology, and financial services. |                                          |                    |
|         |                            | 1.5 Build resilience of the poor and reduce exposure to climate-related and other shocks.                           |                                          |                    |
|         |                            | 1.a Mobilize resources to end poverty.                                                                              |                                          |                    |
|         |                            | 1.b Create pro-poor policy frameworks.                                                                              |                                          |                    |
| 2       | Zero Hunger                | 2.1 End hunger and ensure access to safe, nutritious food year-round.                                               |                                          |                    |
|         |                            | 2.2 End all forms of malnutrition.                                                                                  |                                          |                    |
|         |                            | 2.3 Double agricultural productivity and incomes of small-scale producers.                                          |                                          |                    |
|         |                            | 2.4 Ensure sustainable food production systems and resilient agricultural practices.                                |                                          |                    |
|         |                            | 2.5 Maintain genetic diversity of seeds, plants, and animals.                                                       |                                          |                    |
|         |                            | 2.a Increase investment in rural infrastructure, research, and technology.                                          |                                          |                    |
|         |                            | 2.b Correct and prevent trade restrictions/distortions in global food markets.                                      |                                          |                    |
| 3       | Good Health and Well-Being | 2.c Adopt measures to ensure proper functioning of food commodity markets.                                          |                                          |                    |
|         |                            | 3.1 Reduce global maternal mortality ratio.                                                                         |                                          |                    |
|         |                            | 3.2 End preventable deaths of newborns and under-5 children.                                                        |                                          |                    |
|         |                            | 3.3 End epidemics of AIDS, tuberculosis, malaria, and neglected tropical diseases.                                  |                                          |                    |
|         |                            | 3.4 Reduce premature mortality from NCDs and promote mental health.                                                 |                                          |                    |
|         |                            | 3.5 Strengthen prevention and treatment of substance abuse.                                                         |                                          |                    |
|         |                            | 3.6 Halve global deaths/injuries from road traffic accidents.                                                       |                                          |                    |
|         |                            | 3.7 Ensure universal access to sexual and reproductive healthcare.                                                  |                                          |                    |
|         |                            | 3.8 Achieve universal health coverage.                                                                              |                                          |                    |
|         |                            | 3.9 Reduce deaths from hazardous chemicals, pollution, and contamination.                                           | √                                        | The research shows |

|   |                   |                                                                                     |   |                                                                                                                        |
|---|-------------------|-------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------|
|   |                   |                                                                                     |   | eco-friendly material choice supports health through reduced waste exposure.                                           |
|   |                   | 3.a Strengthen tobacco control (WHO FCTC).                                          |   |                                                                                                                        |
|   |                   | 3.b Support R&D of vaccines and medicines.                                          |   |                                                                                                                        |
|   |                   | 3.c Increase health financing and workforce.                                        |   |                                                                                                                        |
|   |                   | 3.d Strengthen capacity for early warning and risk management.                      |   |                                                                                                                        |
| 4 | Quality Education | 4.1 Ensure all complete free, equitable, quality primary and secondary education.   |   |                                                                                                                        |
|   |                   | 4.2 Ensure access to quality early childhood development and pre-primary education. |   |                                                                                                                        |
|   |                   | 4.3 Ensure equal access to affordable technical, vocational, and higher education.  |   |                                                                                                                        |
|   |                   | 4.4 Increase skills for employment and entrepreneurship.                            | √ | The research builds advanced modeling and simulation skills relevant for engineering education and sustainable design. |
|   |                   | 4.5 Eliminate gender disparities in education.                                      |   |                                                                                                                        |
|   |                   | 4.6 Ensure literacy and numeracy for youth and adults.                              |   |                                                                                                                        |
|   |                   | 4.7 Ensure learners acquire knowledge/skills for sustainable development.           |   |                                                                                                                        |
|   |                   | 4.a Build and upgrade education facilities that are inclusive and safe.             |   |                                                                                                                        |
|   |                   | 4.b Expand scholarships for developing countries.                                   |   |                                                                                                                        |
|   |                   | 4.c Increase supply of qualified teachers.                                          |   |                                                                                                                        |
| 5 | Gender Equality   | 5.1 End all forms of discrimination against women and girls.                        |   |                                                                                                                        |
|   |                   | 5.2 Eliminate violence against women and girls.                                     |   |                                                                                                                        |
|   |                   | 5.3 Eliminate harmful practices (child, early, forced marriage, FGM).               |   |                                                                                                                        |
|   |                   | 5.4 Recognize and value unpaid care and domestic work.                              |   |                                                                                                                        |
|   |                   | 5.5 Ensure women's participation in leadership and decision-making.                 |   |                                                                                                                        |
|   |                   | 5.6 Ensure universal access to reproductive health and rights.                      |   |                                                                                                                        |
|   |                   | 5.a Undertake reforms to give women equal rights to resources.                      |   |                                                                                                                        |
|   |                   | 5.b Enhance use of enabling technology to empower women.                            |   |                                                                                                                        |

|   |                                 |                                                                                          |   |                                                                                                           |
|---|---------------------------------|------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------|
|   |                                 | 5.c Adopt and strengthen policies and laws for gender equality.                          |   |                                                                                                           |
| 6 | Clean Water and Sanitation      | 6.1 Achieve universal and equitable access to safe drinking water.                       |   |                                                                                                           |
|   |                                 | 6.2 Achieve access to adequate sanitation and hygiene.                                   |   |                                                                                                           |
|   |                                 | 6.3 Improve water quality by reducing pollution.                                         |   |                                                                                                           |
|   |                                 | 6.4 Increase water-use efficiency and sustainable withdrawals.                           |   |                                                                                                           |
|   |                                 | 6.5 Implement integrated water resources management.                                     |   |                                                                                                           |
|   |                                 | 6.6 Protect and restore water-related ecosystems.                                        |   |                                                                                                           |
|   |                                 | 6.a Expand international cooperation in water and sanitation.                            |   |                                                                                                           |
|   |                                 | 6.b Support participation of local communities.                                          |   |                                                                                                           |
| 7 | Affordable and Clean Energy     | 7.1 Ensure universal access to affordable, reliable, modern energy services.             |   |                                                                                                           |
|   |                                 | 7.2 Increase substantially the share of renewable energy.                                |   |                                                                                                           |
|   |                                 | 7.3 Double global rate of improvement in energy efficiency.                              |   |                                                                                                           |
|   |                                 | 7.a Enhance international cooperation on clean energy research/technology.               |   |                                                                                                           |
|   |                                 | 7.b Expand infrastructure and upgrade technology for sustainable energy.                 |   |                                                                                                           |
| 8 | Decent Work and Economic Growth | 8.1 Sustain per capita economic growth.                                                  |   |                                                                                                           |
|   |                                 | 8.2 Achieve higher levels of productivity through diversification, tech, and innovation. | √ | The research promotes development of lightweight, high-performance materials for manufacturing industries |
|   |                                 | 8.3 Promote policies for decent job creation and entrepreneurship.                       |   |                                                                                                           |
|   |                                 | 8.4 Improve resource efficiency in production and consumption.                           |   |                                                                                                           |
|   |                                 | 8.5 Achieve full and productive employment for all.                                      |   |                                                                                                           |
|   |                                 | 8.6 Substantially reduce youth not in employment/education/training.                     |   |                                                                                                           |
|   |                                 | 8.7 Eradicate forced labour, modern slavery, and child labour.                           |   |                                                                                                           |
|   |                                 | 8.8 Protect labour rights and safe working environments.                                 |   |                                                                                                           |
|   |                                 | 8.9 Promote sustainable tourism.                                                         |   |                                                                                                           |
|   |                                 | 8.a Increase aid for trade support.                                                      |   |                                                                                                           |
|   |                                 | 8.b Develop a global youth employment strategy.                                          |   |                                                                                                           |

|    |                                          |                                                                                      |   |                                                                                  |
|----|------------------------------------------|--------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------|
| 9  | Industry, Innovation, and Infrastructure | 9.1 Develop quality, reliable, sustainable infrastructure.                           |   |                                                                                  |
|    |                                          | 9.2 Promote inclusive and sustainable industrialization.                             |   |                                                                                  |
|    |                                          | 9.3 Increase access of SMEs to financial services and integration into value chains. |   |                                                                                  |
|    |                                          | 9.4 Upgrade infrastructure for sustainability and resource efficiency.               |   |                                                                                  |
|    |                                          | 9.5 Enhance scientific research and technology development.                          | √ | The research demonstrates applied computational modeling and new design insight. |
|    |                                          | 9.a Facilitate sustainable infrastructure in developing countries.                   |   |                                                                                  |
|    |                                          | 9.b Support domestic tech development and value addition.                            |   |                                                                                  |
|    |                                          | 9.c Increase access to ICT and internet.                                             |   |                                                                                  |
| 10 | Reduced Inequalities                     | 10.1 Achieve income growth of bottom 40%.                                            |   |                                                                                  |
|    |                                          | 10.2 Empower and promote inclusion regardless of status.                             |   |                                                                                  |
|    |                                          | 10.3 Ensure equal opportunity and reduce inequalities of outcome.                    |   |                                                                                  |
|    |                                          | 10.4 Adopt policies for fiscal, wage, and social protection equality.                |   |                                                                                  |
|    |                                          | 10.5 Improve regulation of global financial markets.                                 |   |                                                                                  |
|    |                                          | 10.6 Ensure enhanced representation in global institutions.                          |   |                                                                                  |
|    |                                          | 10.7 Facilitate safe, regular, and responsible migration.                            |   |                                                                                  |
|    |                                          | 10.a Implement special treatment for developing countries.                           |   |                                                                                  |
|    |                                          | 10.b Encourage development assistance and investment in least developed areas.       |   |                                                                                  |
|    |                                          | 10.c Reduce remittance costs.                                                        |   |                                                                                  |
|    |                                          |                                                                                      |   |                                                                                  |
| 11 | Sustainable Cities and Communities       | 11.1 Ensure access to adequate, safe, and affordable housing.                        |   |                                                                                  |
|    |                                          | 11.2 Provide sustainable transport systems.                                          |   |                                                                                  |
|    |                                          | 11.3 Enhance inclusive urbanization and capacity for planning.                       |   |                                                                                  |
|    |                                          | 11.4 Protect cultural and natural heritage.                                          |   |                                                                                  |
|    |                                          | 11.5 Reduce disaster impact and losses.                                              |   |                                                                                  |
|    |                                          | 11.6 Reduce environmental impact of cities (air quality, waste).                     |   |                                                                                  |
|    |                                          | 11.7 Provide access to safe, inclusive green/public spaces.                          |   |                                                                                  |
|    |                                          | 11.a Support positive links between urban, peri-urban, rural.                        |   |                                                                                  |
|    |                                          | 11.b Increase disaster risk reduction strategies.                                    |   |                                                                                  |

|    |                                        |                                                                              |   |                                                                                                                             |
|----|----------------------------------------|------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------|
|    |                                        | 11.c Support least developed countries in sustainable building.              |   |                                                                                                                             |
| 12 | Responsible Consumption and Production | 12.1 Implement 10-Year Framework on sustainable consumption/production.      |   |                                                                                                                             |
|    |                                        | 12.2 Achieve sustainable management and use of resources.                    | √ | The research prioritizes the use of TPU structures, which are generally recyclable and durable, reducing material turnover. |
|    |                                        | 12.3 Halve per capita global food waste.                                     |   |                                                                                                                             |
|    |                                        | 12.4 Manage chemicals and waste sustainably.                                 |   |                                                                                                                             |
|    |                                        | 12.5 Substantially reduce waste generation.                                  |   |                                                                                                                             |
|    |                                        | 12.6 Encourage companies to adopt sustainable practices.                     |   |                                                                                                                             |
|    |                                        | 12.7 Promote sustainable public procurement.                                 |   |                                                                                                                             |
|    |                                        | 12.8 Ensure people have relevant information for sustainable development.    |   |                                                                                                                             |
|    |                                        | 12.a Support developing countries' scientific and technological capacity.    |   |                                                                                                                             |
|    |                                        | 12.b Develop tools to monitor sustainable tourism impacts.                   |   |                                                                                                                             |
|    |                                        | 12.c Rationalize inefficient fossil-fuel subsidies.                          |   |                                                                                                                             |
| 13 | Climate Action                         | 13.1 Strengthen resilience and adaptive capacity to climate-related hazards. |   |                                                                                                                             |
|    |                                        | 13.2 Integrate climate measures into national policies.                      |   |                                                                                                                             |
|    |                                        | 13.3 Improve education and awareness on climate change.                      |   |                                                                                                                             |
|    |                                        | 13.a Implement UNFCCC commitments (mobilize \$100 billion annually).         |   |                                                                                                                             |
|    |                                        | 13.b Promote mechanisms for capacity-building in least developed countries.  |   |                                                                                                                             |
| 14 | Life Below Water                       | 14.1 Reduce marine pollution.                                                |   |                                                                                                                             |
|    |                                        | 14.2 Sustainably manage and protect marine ecosystems.                       |   |                                                                                                                             |
|    |                                        | 14.3 Minimize and address ocean acidification.                               |   |                                                                                                                             |
|    |                                        | 14.4 Regulate harvesting and end overfishing.                                |   |                                                                                                                             |
|    |                                        | 14.5 Conserve at least 10% of coastal and marine areas.                      |   |                                                                                                                             |
|    |                                        | 14.6 Prohibit harmful fisheries subsidies.                                   |   |                                                                                                                             |
|    |                                        | 14.7 Increase economic benefits from sustainable marine resources.           |   |                                                                                                                             |
|    |                                        | 14.a Increase scientific knowledge and marine technology transfer.           |   |                                                                                                                             |

|    |                                        |                                                                          |  |  |
|----|----------------------------------------|--------------------------------------------------------------------------|--|--|
|    |                                        | 14.b Provide access for small-scale artisanal fishers.                   |  |  |
|    |                                        | 14.c Implement international law for oceans.                             |  |  |
| 15 | Life on Land                           | 15.1 Conserve terrestrial and freshwater ecosystems.                     |  |  |
|    |                                        | 15.2 Promote sustainable management of forests.                          |  |  |
|    |                                        | 15.3 Combat desertification and restore degraded land.                   |  |  |
|    |                                        | 15.4 Ensure conservation of mountain ecosystems.                         |  |  |
|    |                                        | 15.5 Take urgent action to reduce biodiversity loss.                     |  |  |
|    |                                        | 15.6 Promote fair benefit-sharing from genetic resources.                |  |  |
|    |                                        | 15.7 End poaching and trafficking of protected species.                  |  |  |
|    |                                        | 15.8 Prevent introduction of invasive alien species.                     |  |  |
|    |                                        | 15.9 Integrate ecosystem values into policies/planning.                  |  |  |
|    |                                        | 15.a Mobilize resources for biodiversity.                                |  |  |
|    |                                        | 15.b Finance sustainable forest management.                              |  |  |
|    |                                        | 15.c Support local communities for forest and wildlife.                  |  |  |
| 16 | Peace, Justice and Strong Institutions | 16.1 Reduce violence and related death rates.                            |  |  |
|    |                                        | 16.2 End abuse, trafficking, and violence against children.              |  |  |
|    |                                        | 16.3 Promote rules of law and equal access to justice.                   |  |  |
|    |                                        | 16.4 Reduce illicit financial/arms flows, organized crime.               |  |  |
|    |                                        | 16.5 Reduce corruption and bribery.                                      |  |  |
|    |                                        | 16.6 Develop effective, accountable institutions.                        |  |  |
|    |                                        | 16.7 Ensure inclusive, participatory decision-making.                    |  |  |
|    |                                        | 16.8 Broaden participation of developing countries in global governance. |  |  |
|    |                                        | 16.9 Provide legal identity for all (including birth registration).      |  |  |
|    |                                        | 16.10 Ensure access to information and protect freedoms.                 |  |  |
|    |                                        | 16.a Strengthen national institutions for prevention of violence.        |  |  |
|    |                                        | 16.b Promote/enforce non-discriminatory laws and policies.               |  |  |
| 17 | Partnerships for the Goals             | 17.1 Strengthen domestic resource mobilization.                          |  |  |
|    |                                        | 17.2 Developed countries to implement ODA commitments.                   |  |  |
|    |                                        | 17.3 Mobilize additional financial resources.                            |  |  |

|  |  |                                                                         |   |                                                                                                 |
|--|--|-------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------|
|  |  | 17.4 Assist developing countries with debt sustainability.              |   |                                                                                                 |
|  |  | 17.5 Invest in least developed countries.                               |   |                                                                                                 |
|  |  | 17.6 Enhance access to science, technology, innovation.                 | √ | The research integrates computational simulation, materials engineering, and design innovation. |
|  |  | 17.7 Promote environmentally sound technologies.                        |   |                                                                                                 |
|  |  | 17.8 Fully operationalize technology bank for LDCs.                     |   |                                                                                                 |
|  |  | 17.9 Enhance international support for capacity-building.               |   |                                                                                                 |
|  |  | 17.10 Promote a universal, rules-based trading system (WTO).            |   |                                                                                                 |
|  |  | 17.11 Increase exports of developing countries.                         |   |                                                                                                 |
|  |  | 17.12 Timely implementation of duty-free, quota-free market access.     |   |                                                                                                 |
|  |  | 17.13 Enhance global macroeconomic stability.                           |   |                                                                                                 |
|  |  | 17.14 Enhance policy coherence for sustainable development.             |   |                                                                                                 |
|  |  | 17.15 Respect national policy space.                                    |   |                                                                                                 |
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# Abstract

Auxetic structures demonstrate negative Poisson's ratio which provides them with unique sharing, deformation and synclastic properties. These properties make them befitting for uses in cushioning, padding or vibration dampening. One of the auxetic structures that have recently been an area of interest for the researchers is the double arrowhead auxetic structure. It shows unique deformation properties due to its lattice cellular structure and finds its utilization in different fields such as sports safety gear, industrial protective gear, spacecrafts and other engineering sectors. This research work focused on the impact response and the application of the double arrowhead structure in sports and industrial safety gears. The application of the double arrowhead structure in this field has not been explored extensively in existing literature. The compressive properties of cellular double arrowhead structure was validated using Finite Element Analysis on ANSYS. The validated model and the outcomes provided a firm footing for exploring its use in sports and industrial safety. A slice of a double arrowhead structured padding was designed on SolidWorks and its deformation and impact response was simulated on ANSYS, imitating an experimental setup, where a cylindrical impactor weight was used to create the effect of sudden impact. The simulations showed the variations among how a solid TPU padding and a double arrowhead structured padding responded to the impact and stress distribution. These simulations were further corroborated using an experimental setup and the double arrowhead padding was proved superior in terms of uniform stress distribution, impact mitigation and weight minimization. The DAH TPU padding showed significantly lower force transmission in comparison to the solid TPU padding along with a 67.4% weight minimization in the final design.

**Keywords:** Double arrowhead structure, Auxetic structure, Impact response, Finite Element Analysis, Sports safety gear.

## Summary

Auxetic structures have diverse applications in sports gear or personal protective equipment. When an athlete falls while playing sports with a great impact or when an object falls on the body of a worker at an industry, in those moments the difference between a serious injury and walking away safely is affected by properly optimized protective gear or padding. The goal of this study was to optimize the protective paddings by reducing the weight of the padding, ensuring easy manufacturing by using Fused Deposition Modeling and using the geometry of auxetic structures to compare stress and impact behavior. The auxetic structure chosen for this study was the Double Arrowhead Structure which has not been studied effectively yet and so far no impact testing has been conducted on it. A research paper that focused on the compressive properties of double arrowhead structures was validated and the impact test behaviour of a slice taken from a protective padding designed with double arrowhead structure and made of TPU was simulated for impact test analysis. The protective padding slice was simulated and then tested using an experimental setup to study its impact response. During the material selection process, the 3D printability, the comfort for the user and the sustainability of the materials used were prioritized.

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## Nomenclatures and Symbol

| Abbreviation | Meaning                         |
|--------------|---------------------------------|
| DAH          | Double Arrowhead                |
| TPU          | Thermoplastic Polyurethane      |
| ABS          | Acrylonitrile Butadiene Styrene |
| NPR          | Negative Poisson's Ratio        |
| AM           | Additive Manufacturing          |
| FDM          | Fused Deposition Modeling       |
| SLA          | Stereolithography               |
| EVA          | Ethylene-vinyl Acetate          |
| PLA          | Polylactic Acid                 |
| FEA          | Finite Element Analysis         |

### List of Symbols

|          |                   |
|----------|-------------------|
| F        | Force             |
| m        | Mass              |
| a        | Acceleration      |
| k        | Wall length       |
| t        | Wall thickness    |
| W        | Width             |
| $\alpha$ | Angle             |
| R        | Large wall length |

# Chapter 1: Introduction and Literature Review

## 1.1 Background of the Study

In recent years, auxetic structures have been studied extensively in order to meet the increasing demand of lightweight materials and protective paddings in safety gears. Moreover, there were certain limitations in the conventionally used protective padding. Hence, researchers have been working on methods to optimize these gears by improving the load bearing capability and energy dissipation without increasing the weight of the lining or padding. This can be easily accomplished by integrating auxetic structures in the design of these paddings.

With the increase of the use of additive manufacturing, the fabrication of auxetic structures has become easier. It is possible to produce them at lower costs. It allows precise control over intricate geometries and promises fine finishing. The increased availability of Fused Deposition Modeling (FDM) 3D printing technology has facilitated more experimental work to develop complex auxetic structures. However, conventional auxetic materials and their uses have been researched but there is no significant research on the application of double arrowhead auxetic materials in the sports and industrial safety gears.

The sudden impact test response of Double Arrowhead structures (mimicking sports injury or fall impact) remains previously unexplored. The goal of this study was to acquire an understanding of the deformation and energy dissipation behavior of double arrowhead structures under sudden impact when they are applied to sports safety gear or personal protective equipment used in industries. One of the main priorities was also the total weight minimization in comparison to the conventional solid paddings.

As a result of their negative Poisson's ratio, auxetic materials exhibit special mechanical properties. They don't become thinner when they are pulled, but expand sideways, which allows them to better absorb and disperse kinetic energy. This is what makes them ideal for use cases involving the combination of flexibility, comfort and protection. Being one of the more potential auxetic structures, a double arrowhead geometry exhibits better stress distribution and recoverability during dynamic loading. This structure can conform to an

impact force and distribute such a force uniformly through the padding, thus fund spreading in order to reduce localized concentrations of stress that could lead to injury or material breakdown.

With TPU (Thermoplastic Polyurethane) as the matrix of the structures, a high elasticity and resilience can be achieved, which makes it possible to produce these prototypes for real applications in an efficient way. The design of the padding can be tailored by tweaking the cell size, wall thickness and arrowhead angle to cater to mechanical needs for sports or industrial applications.

The main goal of this research is to examine the effectiveness of the double arrowhead auxetic design and its proposed application to replace traditional impact absorbing materials. By using computational modeling and impact testing, we gain insights into the manner in which such structures respond to instantaneous loads. The outcome of this study is anticipated to contribute to the design of lightweight, sturdy and energy dissipating protection gears that can be used comfortably for extended periods while protecting human anatomy, machineries and industrial infrastructures.

### **1.1.1 Double Arrowhead Structure**

A 2D honeycomb re-entrant structure was first presented by Almgren in 1985 and there have been subsequent in-depth studies about it in the following three decades. The mechanical properties of the honeycomb structures have been validated by using different experiments and simulations [1]. However, compared to a conventional re-entrant auxetic structure, a double arrowhead structure is relatively newer and has been a field of keen interest for the researchers.

The Double Arrowhead Structure is a metamaterial specialized by its negative Poisson's ratio. Its triangular geometry helps in distributing stress uniformly and reduces the possibility of failure. The auxetic double arrowhead structure endures the stress perpendicularly under applied loads; the structure compresses inwards and thus it shows commendable stress resistance [2]. The geometry of the DAH in a lattice shows excellent indentation resistance under impact or load.

When the current studies aimed to find a metamaterial that would show excellent energy

absorption properties along with being lightweight and showing synclastic behaviour, Double Arrowhead structure is one of the notable structures that demonstrates similar and superior properties, which makes it a suitable structure for its use as protective cushioning or padding [3].

The double arrowhead structure has been previously tested for some of its mechanical properties such as its compressive and tensile properties and it was observed that under compression it stretched and contracted laterally, because of having negative poisson's ratio due to its geometry [4].

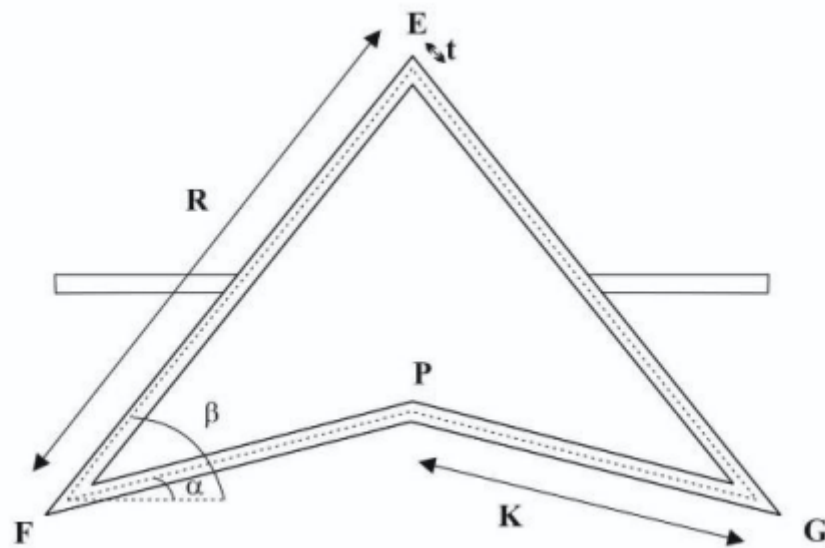


Figure 1.1: Arrowhead shaped unit cell (Adapted from [5]).

A research done by Qiao and Chen shows comparative analyses of the energy dissipation of different auxetic structures such as the re-entrant, chiral structures and the double arrowhead structure and how the latter one outperforms the others in terms of energy dissipation [6].



Figure 1.2: A 2D Double Arrowhead Structure Lattice (Adapted from [6]).

Another similar research by Zhou et al shows that DAH auxetic structure has better specific energy consumption under various loading conditions compared to non-auxetic hexagonal and auxetic re-entrant structures. The DAH structure also displayed superior compressive properties. [7]

### 1.1.2 Additive Manufacturing

Additive Manufacturing is a promising method for producing different products or prototypes in small batches. It is competent in fabricating complex designs or geometries without any material wastage that is found in conventional manufacturing. A hundred percent infill prototype has fine quality and excellent surface finishing. Recent studies have elaborated the advantages of additive manufacturing. Over the past few years, the widespread use of additive manufacturing has caused the costs to go lower, which is advantageous for the manufacturers.

A research work by Parupelli and Desai exhibits the advantage of additive manufacturing in designing complex geometries and the opportunity of customization as per requirement. It also talks about the wide range of applications of additive manufacturing across various

industries [8].

Another study by Balubaaaid and Alsaadi shows the effectiveness of additive manufacturing in material efficiency and waste reduction. The paper focused on how additive manufacturing can be adopted in a way that is sustainable for the government. Furthermore, small batch production can be completed at lower costs with more accuracy and better finishing [9].

Rouf et al, in another work, studied the 3D printing of various industrial products and demonstrated the mechanical advantages of additive manufacturing such as strength and durability and the sustainability aspects [10].

### **1.1.3 TPU Thermoplastic in FDM Printing**

During printing, the TPU filament was extruded through a heated nozzle at 210°C, while the print bed was maintained at 45°C to increase adhesion and reduce warping. A layer height of 0.20 mm was given to maintain surface quality and printing efficiency. Six wall loops were incorporated to enhance structural strength and reduce the likelihood of delamination under mechanical stress. An infill density of 100% with a rectilinear pattern was used to achieve uniform density and improve the load bearing capacity of the double arrowhead structure. Due to the geometry's self sufficiency, there was no requirement for any support structure. An inner and outer brim of 5 mm was applied to improve the adhesion of the printing bed and the “Avoid Crossing Walls” setting was activated to minimize stringing and surface imperfections.

Printing parameters were optimized to accommodate TPU's flexibility. Print speeds for initial and subsequent layers were set to 30 mm/s, while a higher speed of 280 mm/s was used for infill and internal solid infill to reduce overall print duration without compromising quality. To preserve dimensional precision, 5000 mm/s<sup>2</sup> acceleration and a travel speed of 400 mm/s were used. The maximum volumetric flow rate and the retraction length were respectively set to 2.5 mm<sup>3</sup>/s and 3 mm for reduced filament leaking during transit to maintain process consistency.

In order to reduce anisotropy and encourage consistent deformation during testing, the

model's orientation was selected to align the struts and arrowhead units with the printer build direction. This strategy guaranteed precise printing of the auxetic geometry and a robust interlayer bond.

| Parameter                      | Value/Setting                    |
|--------------------------------|----------------------------------|
| Printing Technology            | Fused Deposition Modeling (FDM)  |
| Printing Material              | Thermoplastic Polyurethane (TPU) |
| Nozzle Temperature             | 210 °C                           |
| Bed Temperature                | 45 °C                            |
| Layer Height                   | 0.20 mm                          |
| Wall Loops                     | 6                                |
| Infill Density                 | 100 %                            |
| Infill Pattern                 | Rectilinear                      |
| Support                        | None                             |
| Brim Type                      | Inner and Outer (5 mm)           |
| Retraction Length              | 3 mm                             |
| Max Volumetric Flow            | 2.5 mm <sup>3</sup> /s           |
| Initial Layer Speed            | 30 mm/s                          |
| Outer/Inner Wall Speed         | 30 mm/s                          |
| Internal Solid Infill Speed    | 280 mm/s                         |
| Sparse Infill Speed            | 280 mm/s                         |
| Top Surface Speed              | 200 mm/s                         |
| Travel Speed                   | 400 mm/s                         |
| Acceleration (Normal Printing) | 5000 mm/s <sup>2</sup>           |

| Parameter             | Value/Setting           |
|-----------------------|-------------------------|
| Acceleration (Travel) | 10000 mm/s <sup>2</sup> |
| Avoid Crossing Walls  | ON                      |
| Overhang Speed        | 30 mm/s                 |
| Vertical Shell Speed  | 80 %                    |

Table 1: TPU printing parameters.

## 1.2 Literature Review

### 1.2.1 Overview and Evolution of Auxetic Structures

Auxetic materials are distinguished by a rare mechanical property called Negative Poisson's Ratio (NPR). Due to this characteristic, upon stretching an auxetic material, it expands perpendicular to the direction of the applied force [11].

This counterintuitive response that exhibited this distinct deformation pattern was first observed in a systematic way by R.S. Lakes in the late 1980s which gave rise to re-entrant foam models. He transformed (open-celled) polyurethane foams into re-entrant geometries using a thermomechanical technique [12].

This discovery has initiated a new field of material science research, where we look at properties of materials not purely based on what they are made out of but also on material geometry and overall configuration. K.E. Evans further developed this idea by producing microporous polyethylene with auxetic behavior—this was the precursor of the first polymer auxetic materials [13].

Evans and others extended the mechanics of NPR materials with a second order classification defining re-entrant geometries, and with Rotating Unit Models as well as providing theoretical background to re-entrant geometries and proposed design strategies [14].

In the 2000's, studies on auxetic lattices were extended to various geometries such as chiral, rotating square, arrowhead and star-shaped [15]. The visualization of deformation patterns were observed in both 2D and 3D lattices using photomechanical and additive manufacturing. The use of computational modeling and finite element simulations in the 2010s made it possible to perform a parametric optimization of auxetic geometries for certain purposes, such as impact absorption, vibration damping or sound control. Ren et al. (2020) introduced three dimensional metamaterials via topology optimization techniques which have tunable poisson's ratios [16]. This discovery offered a design blueprint for absorption personalization, customizable energy and stiffness.

Recently, the fabrication of auxetic materials has faced an escalation because of the advances in additive manufacturing (AM)—particularly Fused Deposition Modeling (FDM), Selective Laser Sintering (SLS), and Stereolithography (SLA). Thanks to these processes, it is possible to produce complex auxetic geometries at multiple scales. These can be integrated seamlessly into protective sports equipment, biomedical implants and aerospace panels [17]. A. Almesmari et al. (2023) used flexible thermoplastics like TPU in order to exhibit the 3D printing of polymeric auxetic metamaterials which demonstrated performance repeatability and scalability [18].

### 1.2.2 Auxetic Structures and Their Applications

There are numerous applications of auxetic structures across different fields. They are used in scaffolds in biomedical engineering. Their shearing properties and negative Poisson's ratio allow them to be used in protective gears. Moreover, they have also found use in cash protection systems due to their excellent energy absorption properties [19].

Auxetic structures also demonstrate synclastic behaviour, which is excellent for preventing injury from high speed impact or fall.

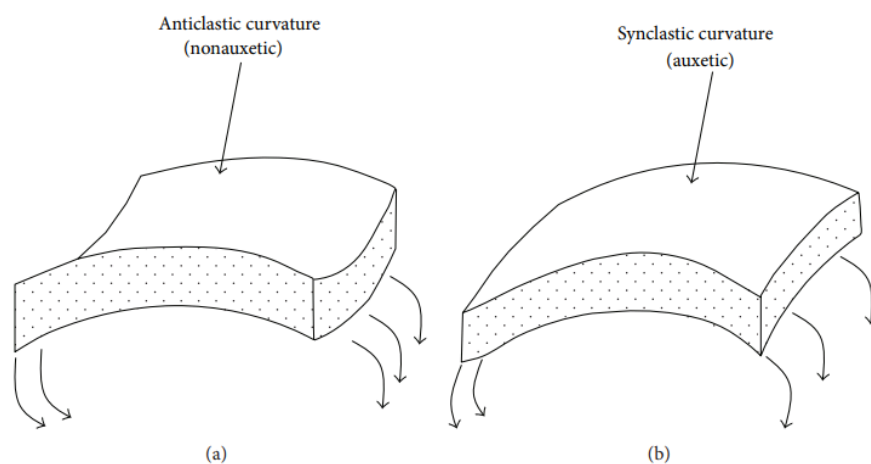


Figure 1.3: (a) Saddle-shaped surface of the conventional material and (b) auxetic dome-shaped doubly curved surface (Adapted from [11]).

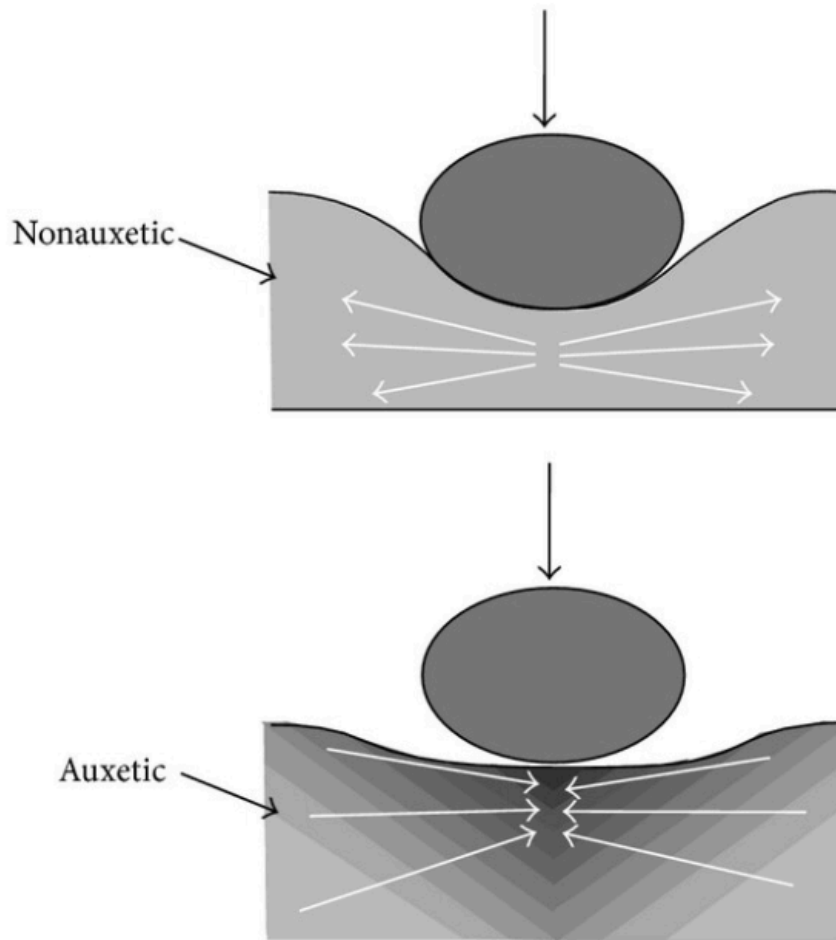


Figure 1.4: Indentation resistance of nonauxetic and auxetic materials (Adapted from [11]).

In this figure, the synclastic behaviour of the auxetic structures are shown. This makes them ideal candidates for not only sport safety gear but also other protective equipment such as bullet proof vests [11]. Auxetic structures are also useful in enhancing worker safety in industrial settings due to their effective stress reduction and vibration absorption [19].

A recent study also explored the application of cylinder-ligament honeycomb in helmets and suggested the use of auxetic structures in sports applications such as, batting pads, batting gloves and football shin pads due to their breathability. [20]



Figure 1.5: Shoe lining made of re-entrant structure (Adapted from [21]).

### 1.2.3 Conventional Protective Gears

In sports safety gears, the foam that is used for cushioning or padding usually undergoes rigorous post processing. These foams are usually low weight, open celled foam. However, their manufacturing is not straightforward. In order to make them suitable for cushioning, post processing steps such as trimming, chemical or heat treatments. They may also require lamination or adhesive bonding. Although conventionally used foams such as EVA foam, polyurethane foam are not costly on their own, these post processing steps ultimately increase the manufacturing costs and time.

A study conducted by Lawrence Livermore National Laboratory (LLNL) material scientists show that 3D printed foams have superior cell topology and better durability in comparison to conventional foams and materials. 3D printed foams have great repeatability and are ideal for bulk production [22].

### 1.2.4 EVA Foam Padding

EVA foam is one of the most widely used materials used in protective padding due to its low density and affordability. EVA foams are generally utilized in protective sports gears such as elbow pads, knee guards and helmet liners. EVA is a co-polymer of vinyl acetate and

ethylene. The amount of vinyl acetate present in the co-polymer determines the durability, flexibility and softness of the EVA foam which typically ranges from 10% to 40% by weight [23].

Eva foams provide a reasonable amount of cushioning via cell wall compression and recovery [24]. However, they have a limited capacity to recover from significant strain deformation which deters them from providing long term protective performance. Under repeated impact loads, the EVA foams lose resilience and durability over time due to creep and permanent deformation [25].

EVA foam exhibits a positive poisson's ratio, for which it contracts laterally under compression. As a result, it gives localized stress concentration instead of distributing it across the surface evenly. Their energy absorption per unit weight is frequently less than that of auxetic metamaterials [26]. The closed cell structure can provide water resistance but also can result in heat accumulation due to limited airflow [27].

### **1.2.5 EVA Foam Manufacturing**

The main manufacturing methods of an EVA foam are-

- Compression molding
- Extrusion
- Injection molding

A mixture of EVA copolymer, blowing agents (like azodicarbonamide), and cross-linking agents (like dicumyl peroxide) are used in compression molding. The copolymer is combined with the agents and heated in a closed mold at 150–190 °C to expand and produce the cellular structure of the foam. The gas released from the decomposition of the blowing agents creates the fine celled structure [28]. The mechanical resilience and morphology is influenced by multiple factors such as the crosslinking temperature, pressure, density and size [29].

Supercritical CO<sub>2</sub> foaming is recent process variation which is used to enhance the environmental performance of the EVA manufacturing process. This process produces microcellular EVA with more uniform and smaller pores with better energy absorption, recyclability, and consistency. It ensures a fine control of nucleation and also eradicates

harmful blowing agents for which the foams show 10%-20% higher energy return efficiency [30].

These studies further prove how rigorous the post-processing of EVA foam is, and these mentioned processes are in fact not environment friendly.

### **1.2.6 TPU Padding**

TPU has been used for cushioning in protective gears for a while now. The application of thin slices of TPU as helmet liner has been studied. It showed how a TPU liner was 10.2 times more effective than a standard helmet lining in absorbing energy in 12 different combinations of neck strength level and location [31].

However, the use of solid TPU padding or cushioning has some limitations, such as - high cost and weight. These can be effectively dealt with by applying double arrowhead structure in the cushioning.

### **1.2.7 Material Selection and Comparison: TPU**

In auxetic and impact-mitigating structures, TPU is highly favorable due to its exceptional combination of fatigue resistance, high elasticity, and 3D printability. TPU can dissipate kinetic energy very efficiently, which makes it ideal for repetitive impact loads [32]. They can undergo large amounts of deformations which are reversible, and can extend by 500%-700% before actually breaking. In contrast to brittle thermoplastics like PLA or ABS, TPU guarantees effective conversion of kinetic energy to thermal energy and permits considerable local deformation without breaking [33].

TPU materials showcase excellent layer adhesion along with its low warping tendency during 3D printing which makes manufacturing intricate, geometrically complicated lattices like the double arrowhead structure possible [34]. The mechanical tunability of TPU can be adjusted by changing the hard-to-soft segment ratio in order to regulate flexibility and stiffness according to the application requirements. So, for adaptive damping effects and protective equipment, TPU is a very adaptable option. [35]

### 1.2.8 Physical Properties of TPU

| Properties                                        | Value      | Unit              |
|---------------------------------------------------|------------|-------------------|
| Density                                           | 1.1–1.3    | g/cm <sup>3</sup> |
| Tensile strength                                  | 20-50      | MPa               |
| Melting temperature                               | 160 to 220 | °C                |
| Glass transition temperature<br>(T <sub>g</sub> ) | -40        | °C                |

Table 2: Physical properties of TPU [35].

Hardness can be made of different grades, from Shore A 60 (soft) to Shore D 80 (stiff) by changing the hard-to-soft segment ratio [33].

TPU has excellent oil resistance and UV stability as well as chemical resistance. Polyether-based TPU has enhanced hydrolysis resistance, which makes it a better fit for humid or hydrolytic environments. Polyester-based TPU showcases higher abrasion resistance and chemical stability for industrial and outdoor environments [33]. This resilience and the viscoelastic properties of TPU make it perfect for applications in protective gear.

### 1.2.9 Mechanical Properties of TPU

TPU is perfect for both soft and semi-rigid lattices because the modulus of TPU can be engineered anywhere between 10 MPa and 150 MPa. It can be adjusted via filler addition or structural composition [37]. TPU mechanically exhibits low hysteresis loss under cyclic stress conditions. This results in most of the absorbed energy recovering elastically with minimum residual strain [38].

In case of 3D printed TPU, most of the mechanical properties are dependent on layer adhesion, infill patterns and density, print speed etc. Overall toughness and anisotropy of the material can be controlled by tweaking these parameters. For instance, higher infill density will result in higher stiffness but lower flexibility [39].

Recent research highlights TPU's compatibility with topology-optimized auxetic structures. When compared to uniform structures (solid foams), the TPU structures made via FDM can improve the overall energy absorption by up to 50%. [40]

#### **1.2.10 Environmental Impact of TPU**

TPU is one of the most useful polymers due to it being known for having high ductility and good biocompatibility [41]. The mechanical properties of TPU makes it desirable for sports gear and protective gear cushioning uses. Nonetheless, using TPU has its challenges because of the way it affects the environment and it is responsible for increasing carbon footprint. Most of the time, waste TPU takes a long time to break down. A study conducted by Vladislav Jašo et al showed the biodegradation process of TPU where it was observed that although TPU starts to degrade very fast initially, its degradation rate becomes slower over time [42]. However, a study published in August 2023, explored different methods to deal with waste TPU by recycling them and reforming them for mine thermal insulation research [43].

Although the TPU filament used to manufacture the prototypes for the testing purposes of this study was not biodegradable, replacing it with biodegradable TPU would be beneficial for the environment in the long run [44].

#### **1.2.11 Printing Technology: Fused Deposition Modelling (FDM)**

Fused Deposition Modeling (FDM) was used to fabricate both the double arrowhead auxetic structure and the tensile and compressive test specimens. This method was chosen for its ability to accurately reproduce complex geometries, accessibility, and compatibility with flexible thermoplastics such as Thermoplastic Polyurethane (TPU). The use of TPU filament is advantageous due to its excellent elasticity, resilience and energy-absorption characteristics; which are particularly beneficial for characterizing the mechanical performance of auxetic and impact-absorbing structures [45]. To ensure consistency in material properties and to allow direct comparison, the TPU was used to fabricate both the solid and the DAH specimens.

### 1.3 Problem Identification and Problem Statement

Traditional impact absorbing materials like foams and rubbers often cannot provide flexibility, weight reduction and effective energy dissipation altogether. Auxetic structures, which have a negative Poisson's ratio, address this issue by becoming denser under pressure, leading to better impact absorption and resilience. When made with a flexible material such as Thermoplastic Polyurethane (TPU), the double arrowhead auxetic design shows improved recovery after deformation and absorbs more energy than standard designs [46], [47].

However, the mechanical performances like stress-strain behavior and impact absorption of such auxetic patterned structures remain insufficiently explored. The applications of the double arrowhead auxetic structure are not found thoroughly in the current literature. This study focuses on the fabrication and performance analysis of a double arrowhead auxetic structure to evaluate its potential for protective gear, automotive body panels, and industrial impact resistant surfaces; where improved energy mitigation and flexibility is crucial.

### 1.4 Significance of the Study

This study is significant as it not only investigates how the double arrowhead auxetic structure behaves under high velocity or sudden impact but it also offers a perspective into how it will potentially behave as a protective padding, which has not been properly addressed in the existing literature. While the different applications of other metamaterials in sports safety and protective gears have been examined, there is no specific study dedicated to the experimental validation of the double arrowhead auxetic geometry under impact. Moreover, this work provides valuable academic and practical insights for the manufacturers and engineers, supporting the adoption of this geometry for making and designing comfortable protective gear that ensures safety and reduces stress by energy dissipation. It encourages innovation in product design by minimization of the material and the net weight. This research brings new angles for academic purposes as well as provides prospects for practical solutions to real-world problems.

## **Chapter 2: Methodology**

### **2.1 Goals and Objectives**

#### **2.1.1 Aim of The Study**

This research aims to study the sudden impact behaviour of protective gear paddings by designing them using the double arrowhead auxetic structure, optimizing for shock absorption, weight reduction and impact mitigation.

#### **2.1.2 Objectives**

The following are the main goals of this study:

- To fabricate and evaluate the double arrowhead auxetic structure with varying thickness for impact absorption using 3D printing and Finite Element Analysis (FEA) to analyze their deformation behavior and energy absorption capacities.
- To conduct drop impact tests experimentally on the 3D printed auxetic structures to simulate real-world impact conditions, assess their energy absorption and deformation recovery and validate the FEA simulation results between traditionally used solid and double arrowhead structured paddings across various thicknesses.
- To fabricate a performance analysis of the double arrowhead auxetic structure for practical uses such as- knee protection padding or similar structures.

## 2.2 Research Methodology

The research methodology adopted for this study followed a systematic approach.

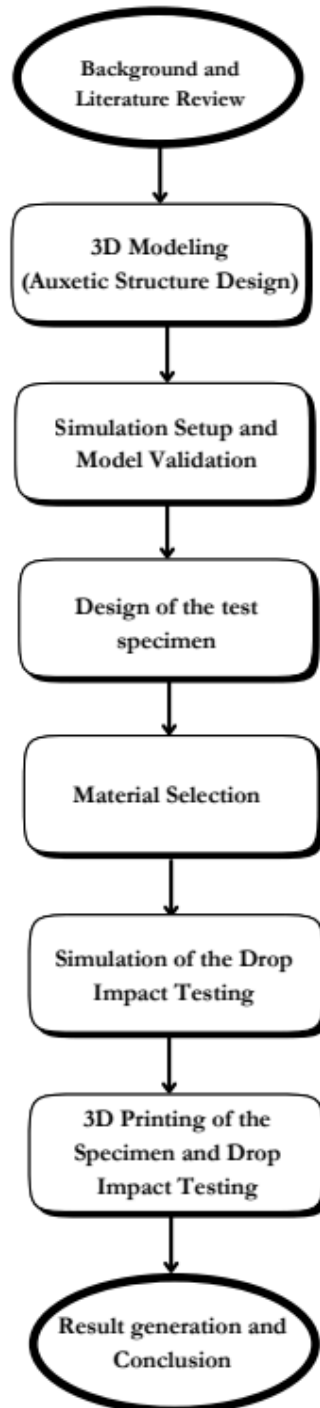


Figure 2.1: Research Methodology Flowchart

To start the research, a comprehensive literature review was done to acquire a general knowledge about the fundamental behavior of auxetic materials and their applications in protective gears. From the prior research on arrowhead and re-entrant structures, their geometric parameters influencing negative Poisson's ratio, compression behaviour and stress distribution were analyzed.

Analyzing the insights from previous studies, a double arrowhead unit cell was designed to reduce weight with effective energy absorption. The initial dimensions were taken from Wani and Walame (2021), with a small wall length ( $K$ ) of 20 mm, wall thickness ( $t$ ) of 1.5 mm, width ( $W$ ) of 10 mm, angle ( $\alpha$ ) of  $15^\circ$ , and a large wall length ( $R$ ) of 31.73 mm [5].

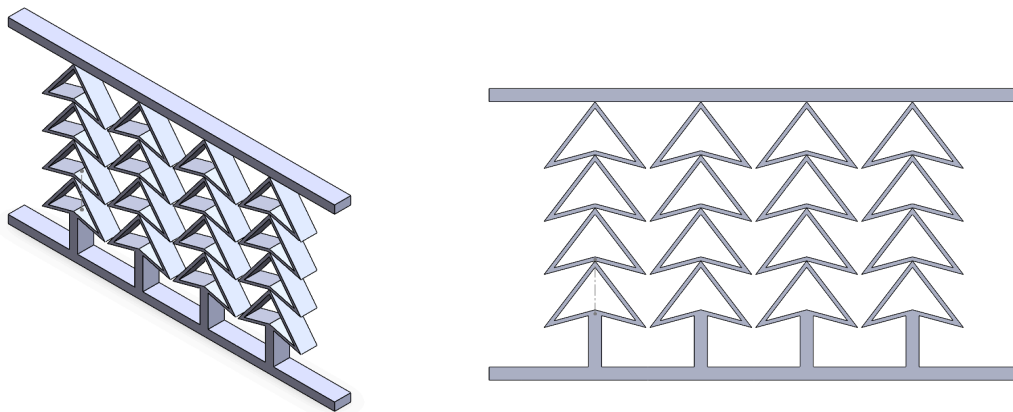


Figure 2.2: Cellular Double Arrowhead Structure for Validation

The mechanical behaviour of the cellular double arrowhead structure, as found in the study done by Wani and Walame, was validated by replicating the simulation on ANSYS [5]. The variation in the outcomes of the comparative study were negligible and hence it gave us the premise for further research.

Following the verification of results, a design concept was modified for real application in shock absorbing paddings implemented in much protective equipment such as helmet, knee guard and shin guard.

Suitable materials were studied based on their mechanical properties and potential applicability in protective equipment. The primary considerations included impact resistance, elastic modulus, density, and energy absorption capacity. Polymers such as thermoplastic polyurethane (TPU) and ABS were shortlisted for their flexibility and durability. After more

consideration and experimentation, TPU was chosen as the selected material.

Finite Element Analysis (FEA) was conducted to simulate the impact and compression loading condition. The model was meshed with tetrahedral elements where the regions of concentrated stress were meshed with fine resolution. Moreover, the boundary condition was applied to the parts as per their real-world constraints and the material property was assigned depending on the selected polymer. The simulation was performed using ANSYS that focused mainly on the stress distribution, deformation and the behavior of energy absorption.

The specimens for the drop impact testing were 3D printed using FDM. To analyze the impact energy absorption performance of the structure, both simulated and experimental drop impact testing was conducted where the structure was allowed to absorb and dissipate energy. The deformation pattern of the double arrowhead and how the loading was dissipated and stress distribution was critically analyzed to determine the redesign improvements.

For the drop impact testing, the accelerometer took the acceleration readings. The mass of the impactor was already known ( 5 kilograms). Since,  $F = ma$ , where F= Force measured in kilonewtons, N and m and a are the mass (kg) and acceleration (m/s<sup>2</sup>) respectively. From the acceleration and the mass, the force, as in, the impact was calculated for each specimen.

To determine how the modified structure performed, the simulation and impact test results were analyzed. The parameters that were compared between the double arrowhead patterns and the traditional solid paddings were specific energy absorption, total deformation, and maximum von Mises stress. The outcomes of the study have been discussed later in detail.

## **Chapter 3: Modeling, Simulation and Physical Experimentation**

### **3.1 Validation of the Compressive Properties of the Double Arrowhead Structure**

The compressive properties of the double arrowhead structure was validated against geometric and mechanical parameters reported in literature to confirm accuracy and reliability. Key validation criteria included dimensional accuracy, auxetic behavior (negative Poisson's ratio), and stress distribution characteristics under compression loading.

Wani and Walame studied the mechanical behavior of cellular arrowhead structure exhaustively and simulated the stress concentration and energy absorption in ANSYS [5]. The study also compared the stress distribution pattern of the double arrowhead structure with the traditional re-entrant honeycomb structure and concluded that the cellular double arrowhead structure showed reduced stress distribution. In order to proceed with this work and study how the double arrowhead structure would respond in a high impact situation, the findings in their study was validated by replicating the exact design of the Double Arrowhead Structure on SolidWorks.

#### **3.1.1 3D Modeling of the Double Arrowhead Structure**

A double arrowhead pattern structure was prepared from a reference, symmetrically, where two arrowhead shapes are joined in such a way that the base of one arrowhead is connected to the tip of the other arrowhead shape placed below. These individual arrowheads are called unit cells. On a larger scale of this lattice, these unit cells are arranged in a repeated pattern; both horizontally and vertically.

The unit cell geometries were designed based on dimensional parameters reported in a previous study. The unit cell dimensions were defined with the following parameters: small wall length ( $K$ ) = 20 mm, wall thickness ( $t$ ) = 1.5 mm, width ( $W$ ) = 10 mm, angle ( $\alpha$ ) = 15 mm, and large wall length ( $R$ ) = 31.73 mm. Using this setup as a singular unit, a validation of

said literature was done with the final experimental geometry being designed using 4 X 4 horizontally and vertically placed cells [5].

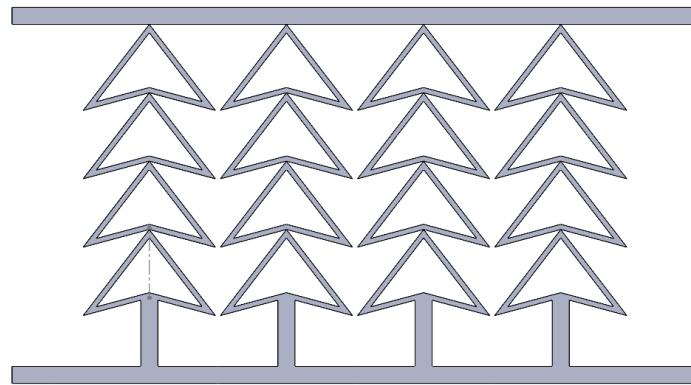


Figure 3.1: Cellular Double Arrowhead Structure for Validation

### 3.1.2 Simulation Results of the Comparative Study

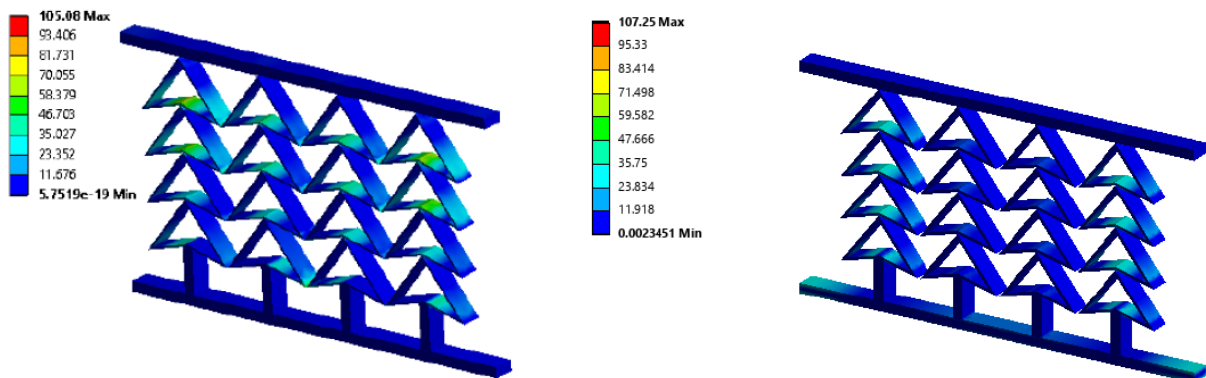


Figure 3.2: Equivalent Stress comparison of Previous Research vs Validation (Linear Case)

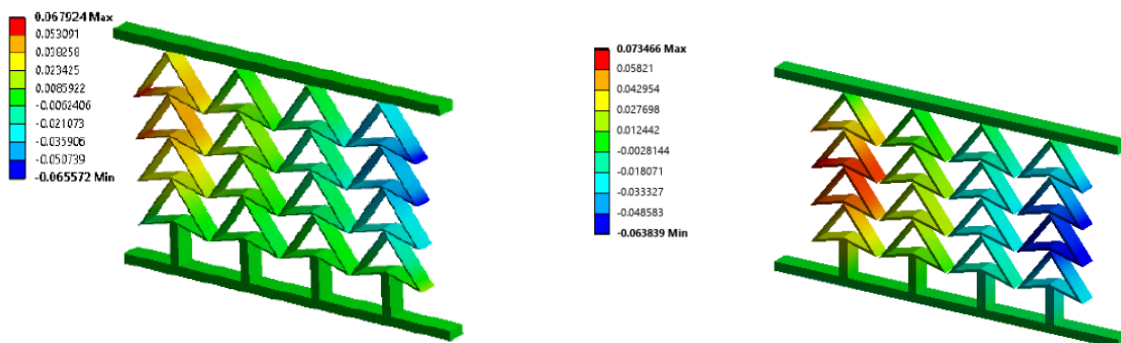


Figure 3.3: Deformation comparison of Previous Research vs Validation (Linear Case)

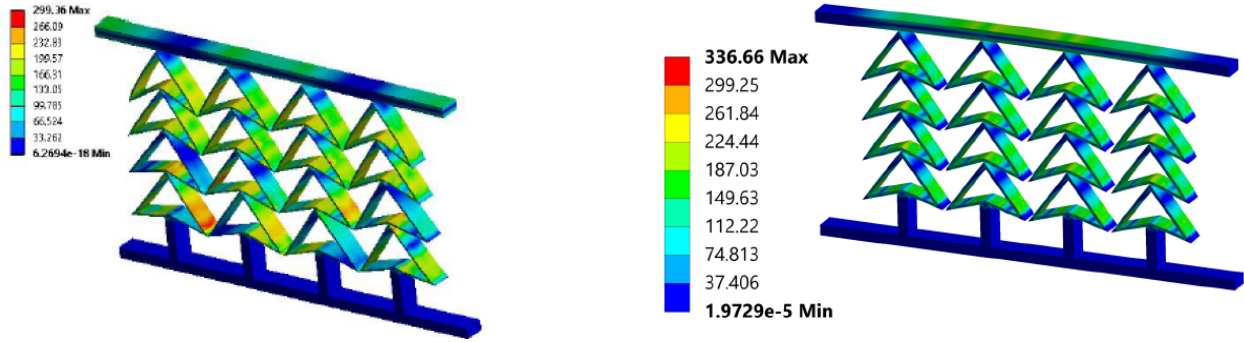


Figure 3.4: Equivalent Stress comparison of Previous Research vs Validation (Non linear case)

| Mechanical Properties                         | Reference Paper   | Validation        | Result Variation |
|-----------------------------------------------|-------------------|-------------------|------------------|
| Maximum Equivalent Stress for Linear Case     | <b>105.08 MPa</b> | <b>107.25 MPa</b> | <b>2.06%</b>     |
| Maximum Deformation for Linear Case           | <b>0.0679 mm</b>  | <b>0.0734 mm</b>  | <b>8.1%</b>      |
| Maximum Equivalent Stress for Non-Linear Case | <b>299.36 MPa</b> | <b>336.66 MPa</b> | <b>12.45%</b>    |

Table 3: Result variation in validative study.

The simulation results closely matched their results. Some of the designing parameters such as the length of the upper and bottom plate and the linkages are absent in the paper. These parameters were assumed, which reduced the possibility of hundred percent error free validation of the original work. However, the percentage of error was within an acceptable range and thus it provided a solid foundation to research the potential applications of this geometry in safety gear.

### 3.2 Simulation of the Drop Impact Testing

Before proceeding with the drop impact testing, simulations were carried out on ANSYS to understand the stress distribution and deformation behavior of the solid TPU padding and the DAH padding under impact and to understand how they would behave in a real-life scenario that would implicate a high-impact injury or fall.

### 3.2.1 Design of the Test Specimen

After validating the initial results, the dimensions of the structure were modified to align with real-life impact-absorbing paddings used in protective equipment such as helmets, knee guards, shin guards, and back protectors. In such applications, the padding thickness typically ranges from 5 mm to 8 mm [48]. Accordingly, the redesigned structure was assigned dimensions of 63 mm in length, 22 mm in width, and 7 mm in thickness. The demonstration model comprised a total of 24 arrowhead units arranged in two rows of 12 unit cells each. This model shall be named Model A.

However, a different CAD model was designed that maintained the dimensions according to the validated model as well. This second model shall be called Model B. The model B was assigned dimensions of 187 mm in length, 200 mm in width, and 60 mm in thickness. The demonstration model comprised a total of 8 arrowhead units arranged in two rows of 4 unit cells each.

The CAD models were designed in SolidWorks as mentioned in the previous chapter.

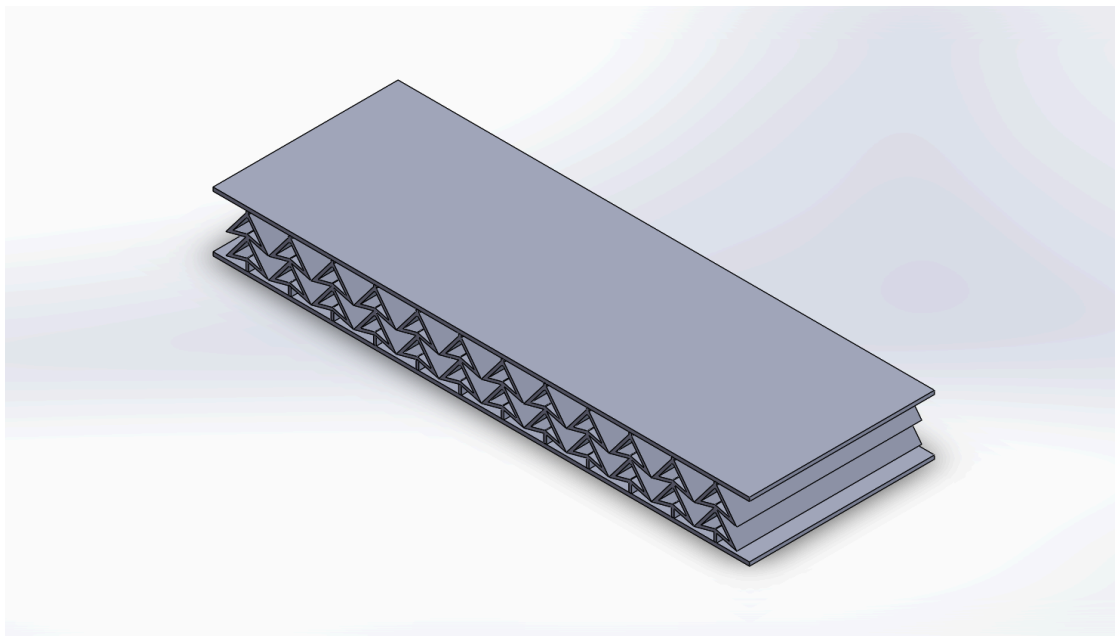


Figure 3.5: Isometric View of the Test Specimen (Model A)

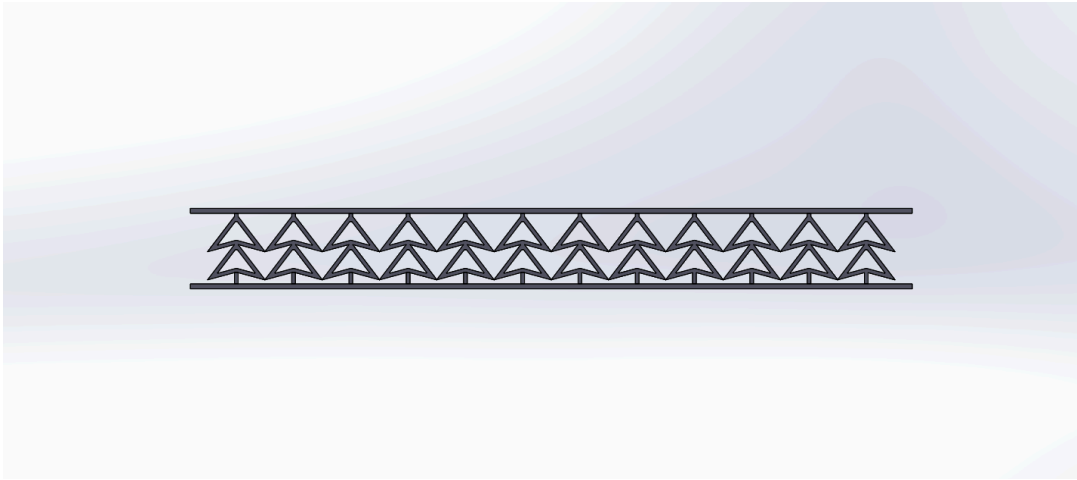


Figure 3.6: Front View of the Test Specimen (Model A)

The design that was done according to the scale of the validated model, Model B is shown below.

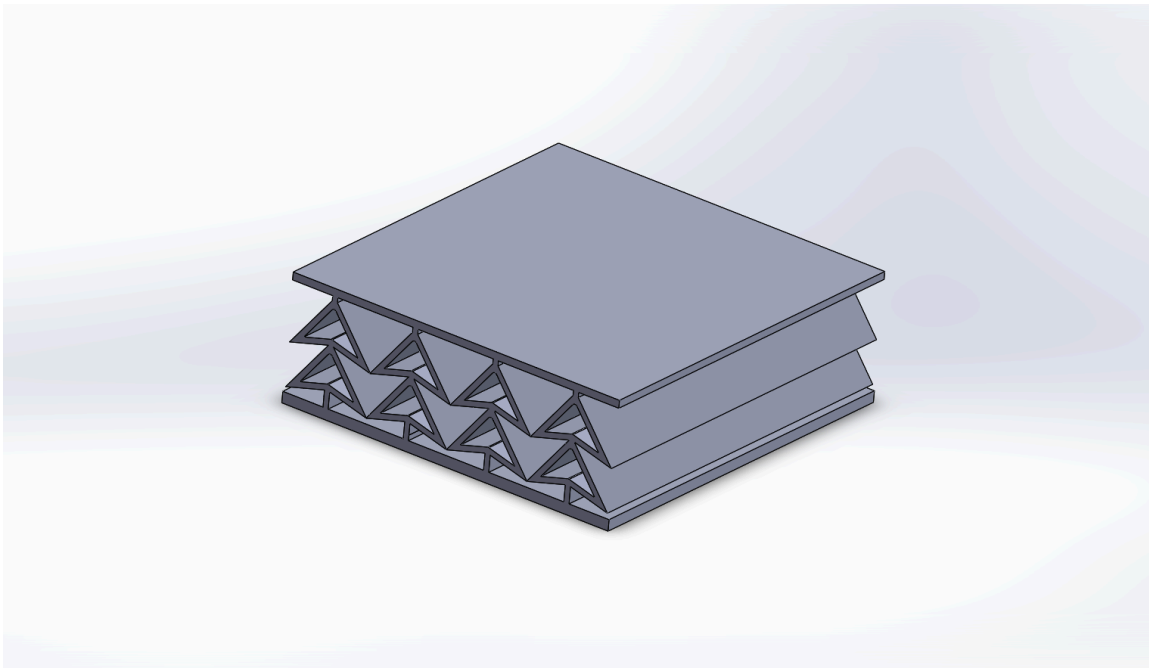


Figure 3.7: Isometric view of the test specimen (Model B)

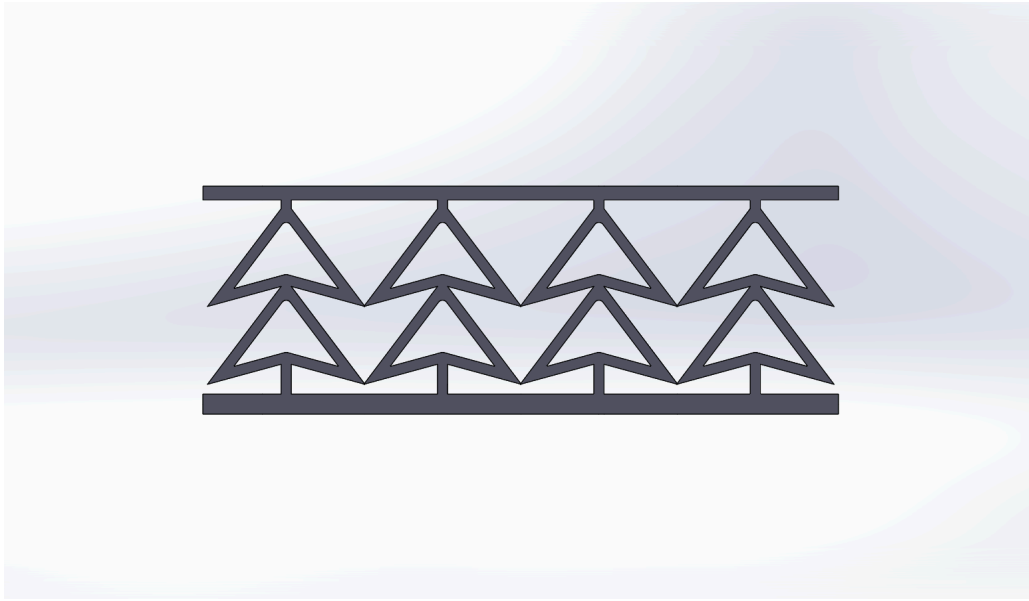


Figure 3.8: Front view of the test specimen (Model B)

Now, two solid test specimens were also designed that would imitate the behaviour of a solid TPU Padding. These models shall be called Model C and Model D respectively. This study also explored how a double arrowhead structure patterned padding behaves in comparison to solid paddings made of the same material. Hence, Model C and Model D were designed corresponding to the exact dimensions of Model A and B respectively.

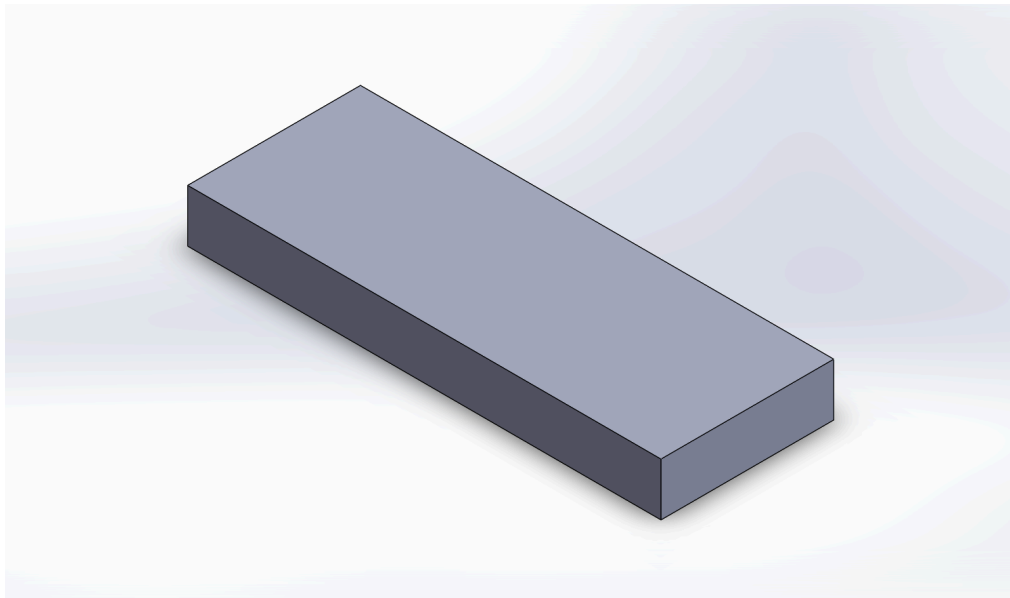


Figure 3.9: Solid test specimen (Model C)

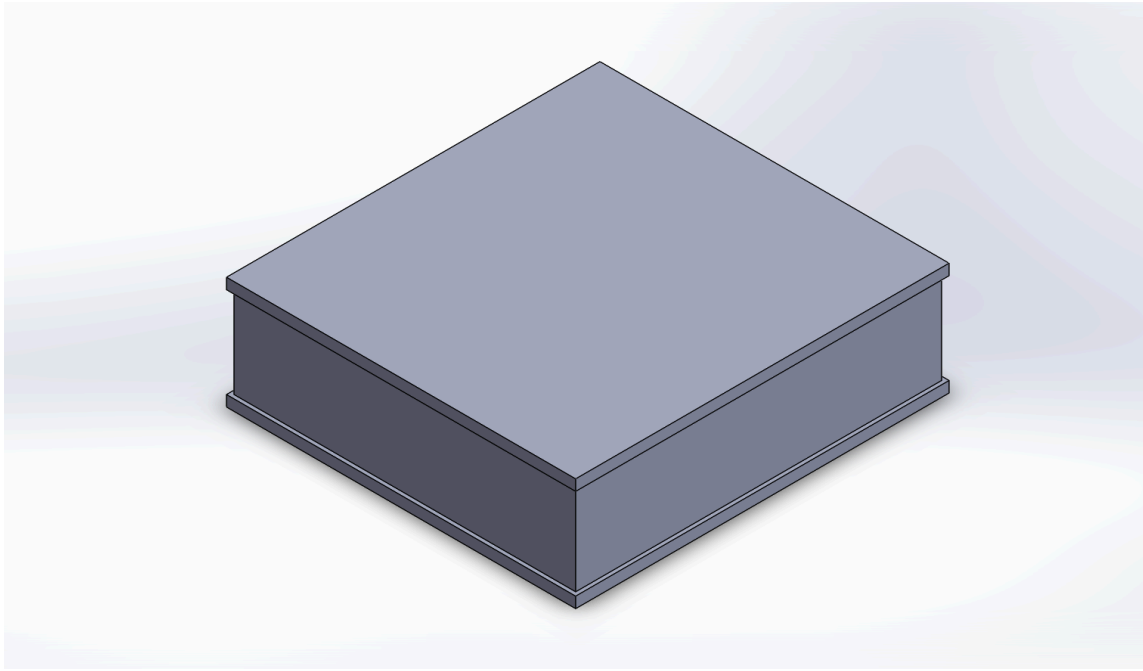


Figure 3.10: Solid test specimen (Model D)

The cylindrical impactor that was used to simulate the impact loading was also modeled. It was assigned a diameter of 132 mm and a mass of 4.976 kilograms.

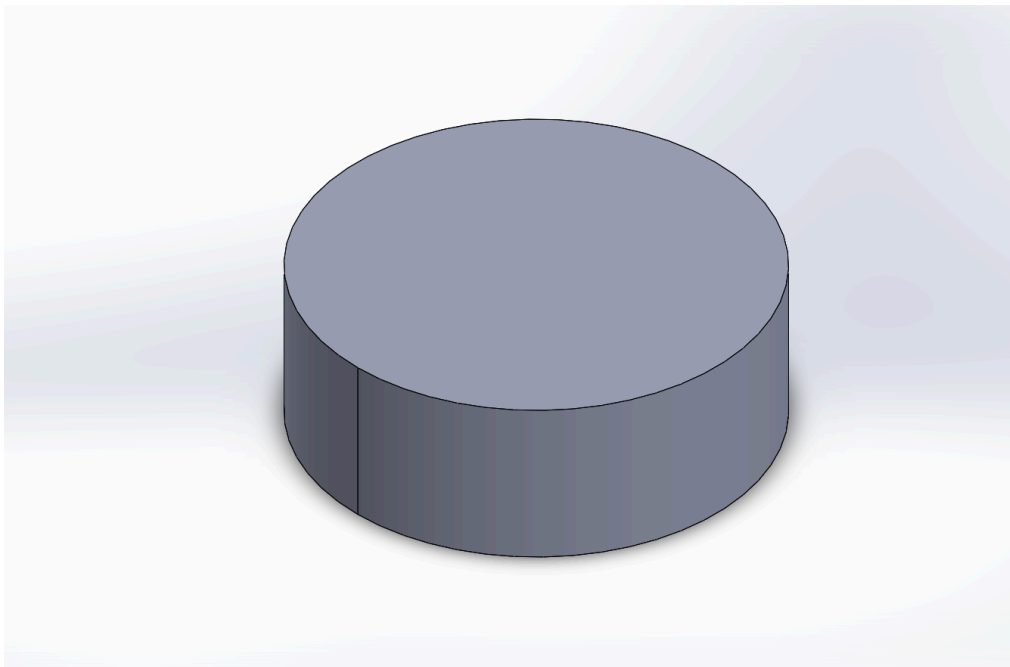


Figure 3.11: Cylindrical Impactor

### 3.2.2 Simulation Setup

The models were then imported to ANSYS Workbench and simulated using the following boundary conditions.

The lower surface of the test specimens were provided a **fixed support**. Since there will be a body surface beneath the protective padding, it was checked how much stress is possible to transmit via the protective padding. The velocity of the cylindrical impactor weight with which it will strike the protective padding was taken **10 m/s**. The cylindrical impactor weight had a mass of **4.976 kilograms**, which is almost equal to the impactor weight used in the experimental setup which is 5 kilograms.

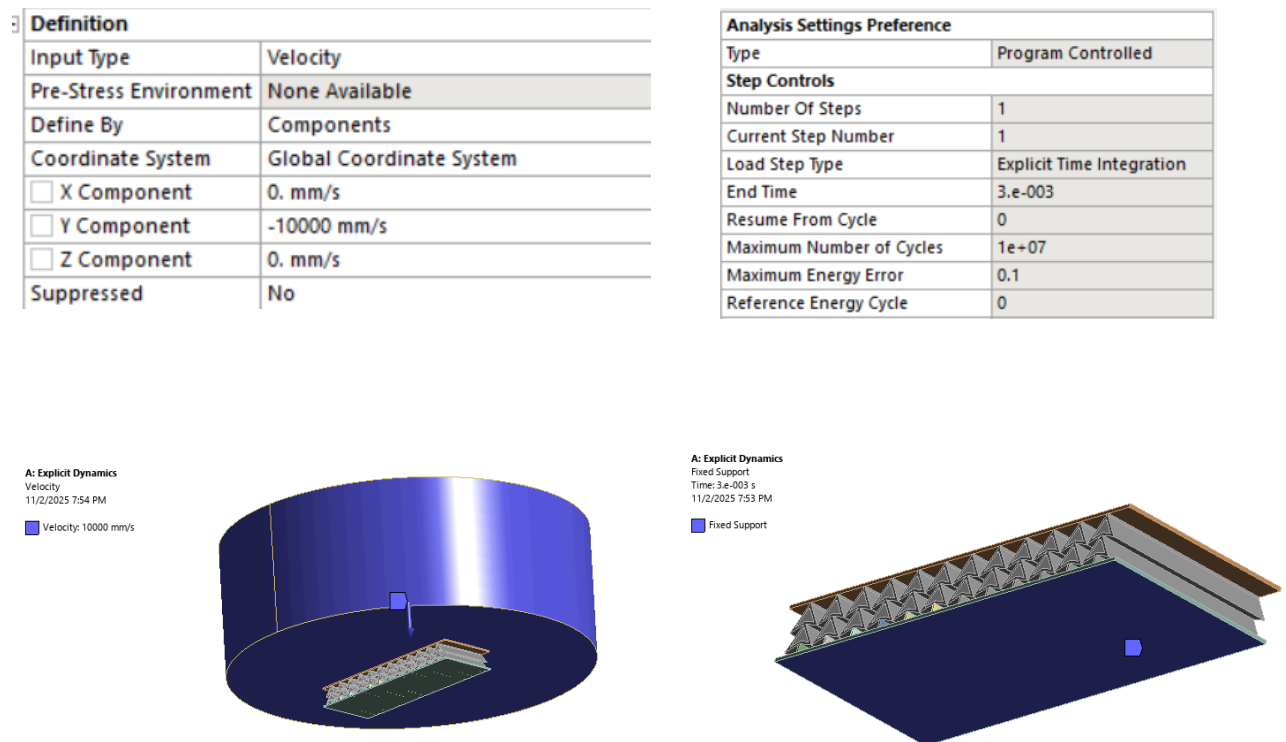


Figure 3.12: Boundary Conditions in ANSYS for Model A

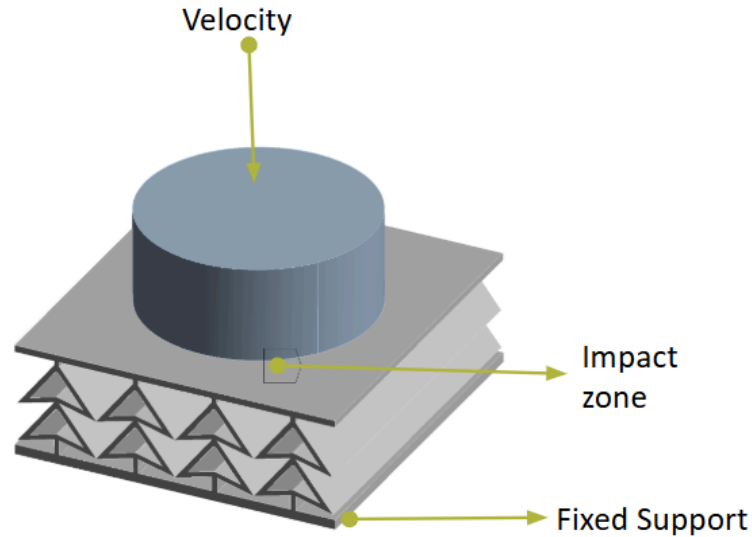


Figure 3.13: ANSYS Setup for Model B

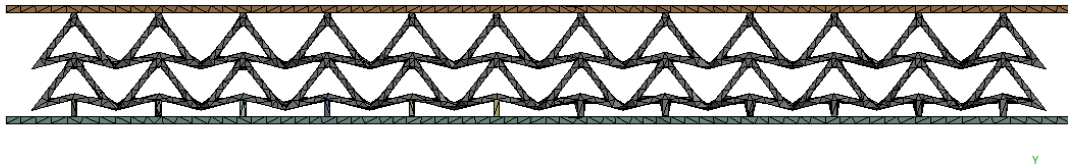


Figure 3.14: Meshing of DAH model

The double arrowhead auxetic structure was meshed using tetrahedral elements to capture its complex geometry. The results stabilized at an element size of 1.4 mm for Model A and C. Hence, a 1.4 mm tetrahedral mesh was used in the final simulation to ensure accuracy while maintaining computational efficiency. Similarly, the final element size for Model B and D was 4 mm.

Four different types of simulation were run, using Model A, Model C, Model B and Model D respectively. In case of Model A, the design thickness of the padding was smaller in order to explore its impact response in applications of protective sports knee guards or shin guards whereas in case of the Model B, the design parameters of the double arrowhead unit cells were kept the same as the research paper that was validated.

### 3.2.3 Simulation Results

#### Simulation 1: Model A- Double Arrowhead (DAH) Padding (7 mm)

The validated double arrowhead structure inside the auxetic model ( $63 \text{ mm} \times 22 \text{ mm} \times 7 \text{ mm}$ ) was subjected to the same impact loading conditions.

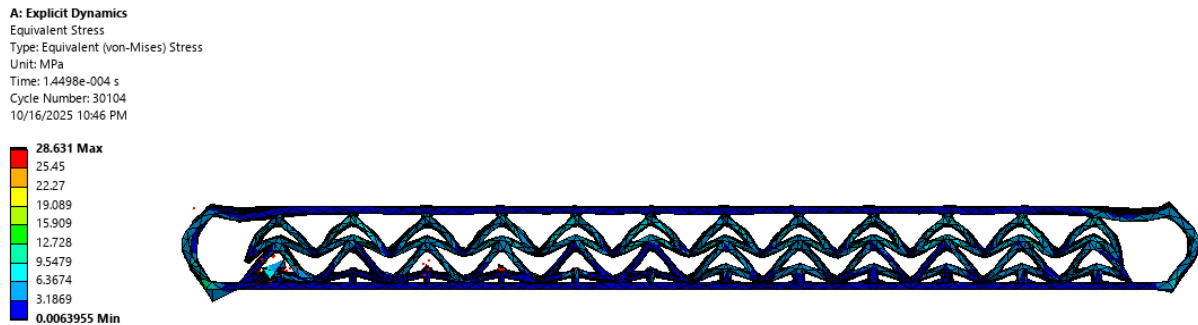


Figure 3.15: Equivalent Von-Misses Stress of Model A (front view)

Upon impact, the upper layer cell walls rotated inward, resulting in lateral expansion instead of contraction. This showed the desired negative Poisson's ratio (NPR) behaviour of the DAH structure. Stress contour is spread laterally across the unit cells of DAH. The maximum Von Mises stress was recorded as 28.631MPa, which is 40-45% lower than the solid TPU. The minimum stress on the bottom layer was extremely low, only 0.00639MPa. This means that almost all of the stress was dispersed within the double arrowhead structure and barely reached the bottom layer at all which is connected to the wearer.

**A: Explicit Dynamics**  
 Equivalent Stress  
 Type: Equivalent (von-Mises) Stress  
 Unit: MPa  
 Time: 1.4498e-004 s  
 Cycle Number: 30104  
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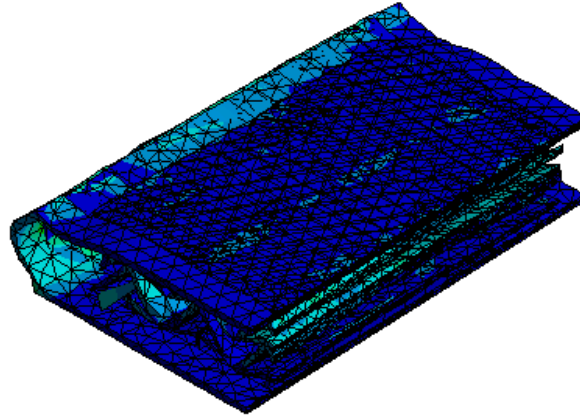
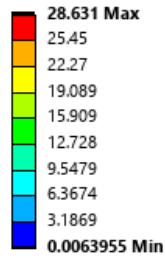


Figure 3.16: Equivalent Von-Misses Stress of Model A (isometric section view)

**A: Explicit Dynamics**  
 Total Deformation  
 Type: Total Deformation  
 Unit: mm  
 Time: 1.4498e-004 s  
 Cycle Number: 30104  
 10/16/2025 10:47 PM

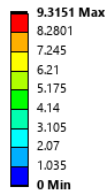


Figure 3.17: Total Deformation of Model A (front view)

**A: Explicit Dynamics**  
 Total Deformation  
 Type: Total Deformation  
 Unit: mm  
 Time: 1.4498e-004 s  
 Cycle Number: 30104  
 10/16/2025 10:53 PM

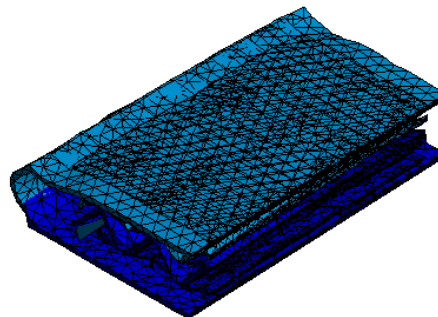
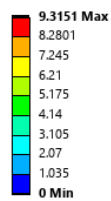


Figure 3.18: Total deformation of Model A (isometric section view)

The simulation result of the specimen with 7 mm thickness shows that the maximum deformation, 9.3151 mm, acted mainly at the top layer of the double arrowhead unit cells near the center of the impact area. This localized deformation indicates that the top layer had absorbed the majority of the applied load due to its direct exposure to the impact surface. The characteristic auxetic geometry of the DAH structure initiated lateral contraction during compression, which contributed to the dissipation of stress within the upper layer. In contrast, the intermediate and bottom layers showed little deformation, since the structure distributed the load in a way that reduced the stress reaching these layers. The bottom layer stayed durable, confirming that most deformation was confined to the top of the specimen.

### Simulation 2: Model C- Solid TPU Padding (7 mm)

In the first simulation, the small solid TPU specimen with similar dimensions to the auxetic model (63 mm × 22 mm × 7 mm). was tested with the cylindrical impactor. The stress distribution of the padding is shown below. In case of the solid TPU padding, the minimum stress transmitted was 0.054 Mpa which is concentrated primarily at the center of the contact region and directly downwards. From the color contour of the stress, it is clear that the bottom covering of the padding is unable to redistribute the stress evenly. Therefore, this simulation showcases higher localized stress of the solid structure of the TPU material.

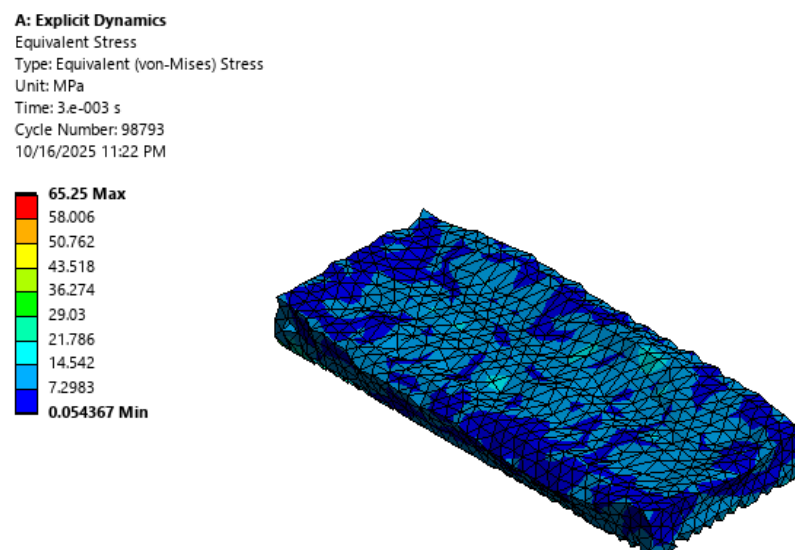


Figure 3.19: Equivalent Von-Misses Stress of Model C (isometric view)

From the deformation pattern, minimal internal energy dissipation is noted. Most of the deflection was concentrated in the impact zone and was transmitted to the base layer almost

directly. The maximum deformation acts upon the uppermost layer and it is measured to be 65.25 mm. But the amount of deformation that occurs in the bottom layer; where it directly impacts the human body, is 0.05 mm. Whereas, in the double arrowhead structure, the amount of deformation in the bottom layer is 0 mm. Thus, the protected area will be more secured and impact free with the application of DAH structure in the paddings instead of solid TPU.

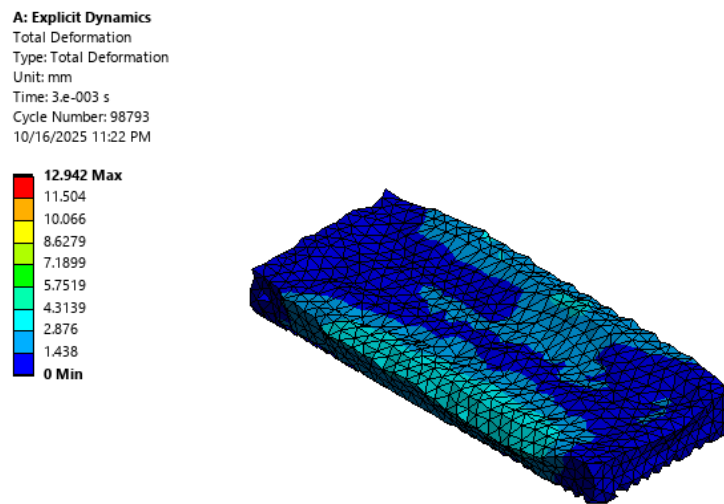


Figure 3.20: Total Deformation of Model C (isometric view)

Since the solid block was compressed downward and also contracted inward, it showed a positive Poisson's ratio response. This behaviour is consistent with the well-known drawbacks of the conventional paddings, where due to the poor, inadequate load distribution, maximum stress is transferred to the wearer. The cushioning partially absorbed some of the impact energy because of the behavioral properties of TPU but it lacked the geometric mechanism necessary to distribute the force in order to protect the body of the wearer.

| Model Name                              | A     | C    |
|-----------------------------------------|-------|------|
| Stress transmitted to inner plate (Mpa) | 0.006 | 0.05 |

Table 4: Inner plate stress comparison of Model A and Model B

### Simulation 3: Model B- Double Arrowhead (DAH) padding (60 mm)

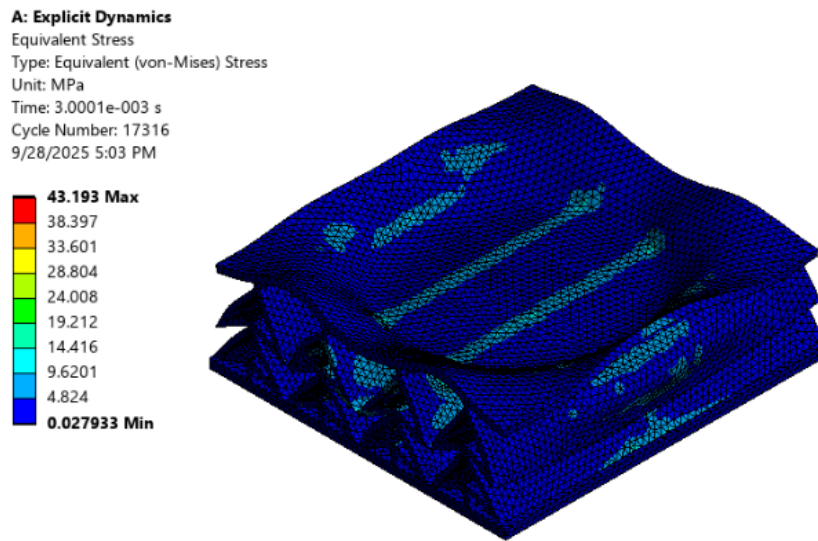


Figure 3.21: Equivalent Von-Misses Stress of Model B (isometric view)

In the case of Model B, it showed excellent stress distribution. Although the stress was much higher compared to the solid TPU block, it eventually distributed the stress efficiently and the stress distributed to the lower plate was 0.02 Mpa which is much lower than the stress transmitted to the bottom plate of the solid TPU block (0.05 Mpa). The triangular unit cells of the double arrowhead structure helps to distribute the stress throughout the whole body without passing down the stress directly to the bottom plate. This makes the structure suitable for safety uses compared to the solid TPU block as it would indicate that the body part in contact with the bottom plate would not be subjected to high impacts.

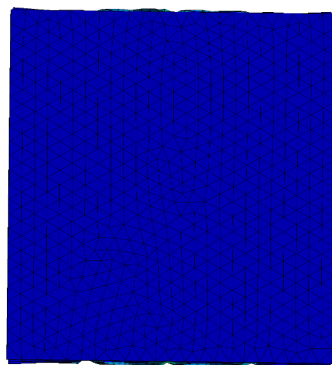


Figure 3.22: Equivalent Von-Misses Stress on the bottom plate of Model

The bottom plate shows a minimum stress transmission of 0.02 Mpa and throughout the bottom plate the stress does not vary at all.

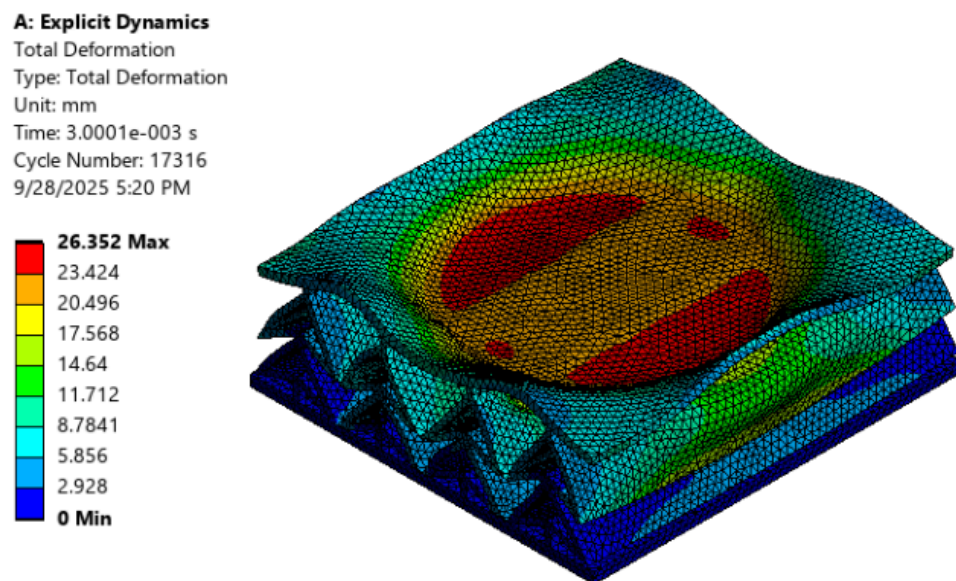


Figure 3.23: Total Deformation of Model B (isometric view)

The maximum total deformation reported was 26.352 mm, while the minimum deformation was 0 mm; indicating that the displacement was efficiently absorbed and dissipated across the structure. The arrowhead shaped unit cell design allows the structure to deform flexibly rather than transmitting the displacement directly towards the bottom plate. This deformation behavior makes the structure highly suitable for impact absorption and safety application since it suggests that the surface or body part in contact with the bottom plate would experience minimal movement and reduced impact intensity.

#### Simulation 4 : Model D- Solid TPU Padding (60 mm)

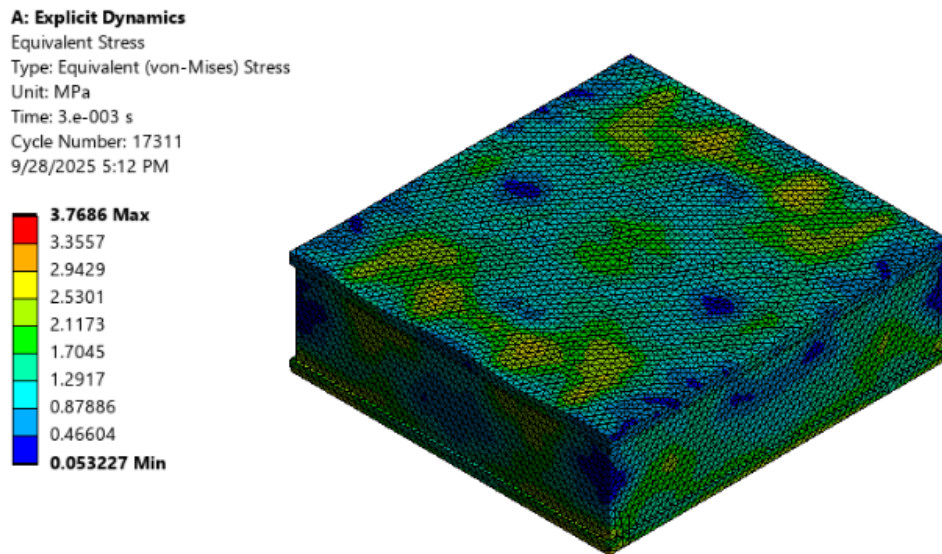


Figure 3.24: Equivalent Von-Mises Stress of Model D (isometric view)

In this simulation, where a larger scale solid TPU block was subjected to the load of the cylindrical impactor, the solid block showed a stress of 3.76 Mpa. The TPU block has a higher weight compared to the DAH structure which is why the resulting stress was very low. However, stress distribution was not uniform and stress was distributed to the bottom plate as well. The minimum stress transmitted to the bottom was 0.05 Mpa.

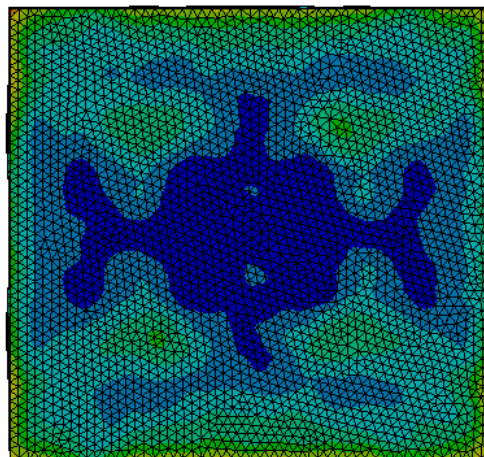


Figure 3.25: Equivalent Von-Mises Stress on the bottom plate of Model D

The bottom surface undergoes massive stress transmission that ranges from 0.02 Mpa (in the centre) up to 2.1 Mpa. An impact of 2.1 Mpa is extremely dangerous to the human body thus proving that solid TPU paddings or liners are not safe enough as protective paddings.

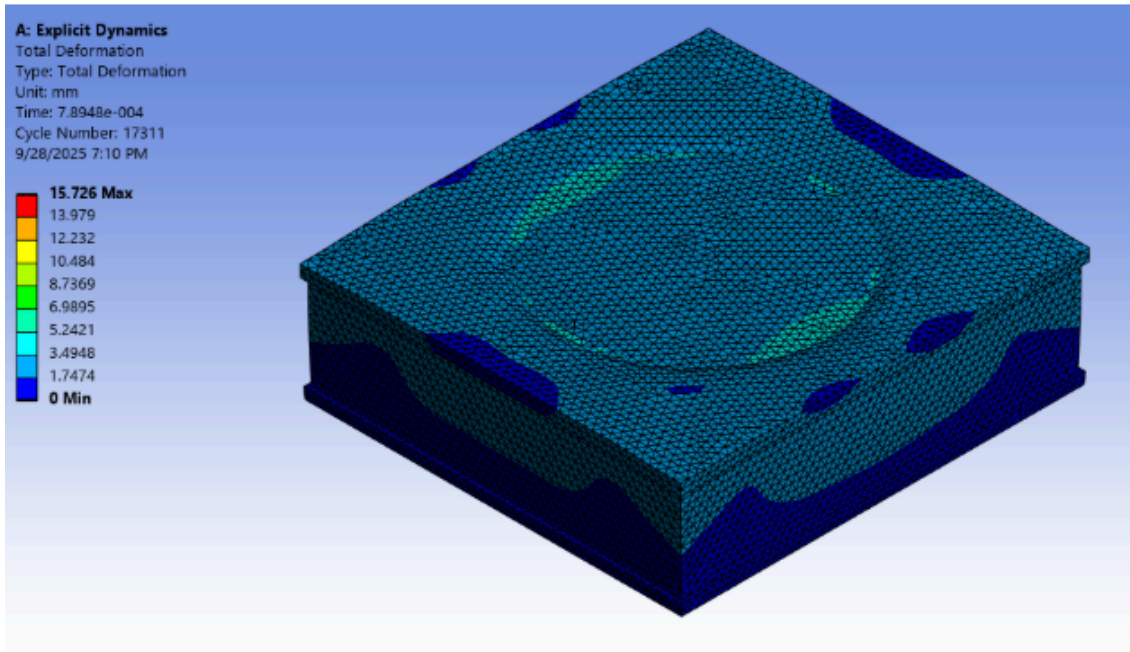


Figure 3.26: Total Deformation of Model D (isometric view)

In case of deformation, the maximum stress was 15.726 mm and while the deformation was not high, it does not demonstrate synclastic behaviour like auxetic structures, leading to inefficient stress distribution throughout the body.

| Model Name                              | B    | D        |
|-----------------------------------------|------|----------|
| Stress transmitted to inner plate (Mpa) | 0.02 | 0.02-2.1 |

Table 5: Inner plate stress comparison of Model B and Model D

### 3.3 Experimental Setup and Procedure

#### 3.3.1 3D Printing and Experimental Setup

To corroborate the ANSYS simulations, the specimens were fabricated using Fused Deposition Modeling and were made of TPU. The test specimen had an upper covering and a lower covering (the lower covering is assumed to be in touch with the human body part).

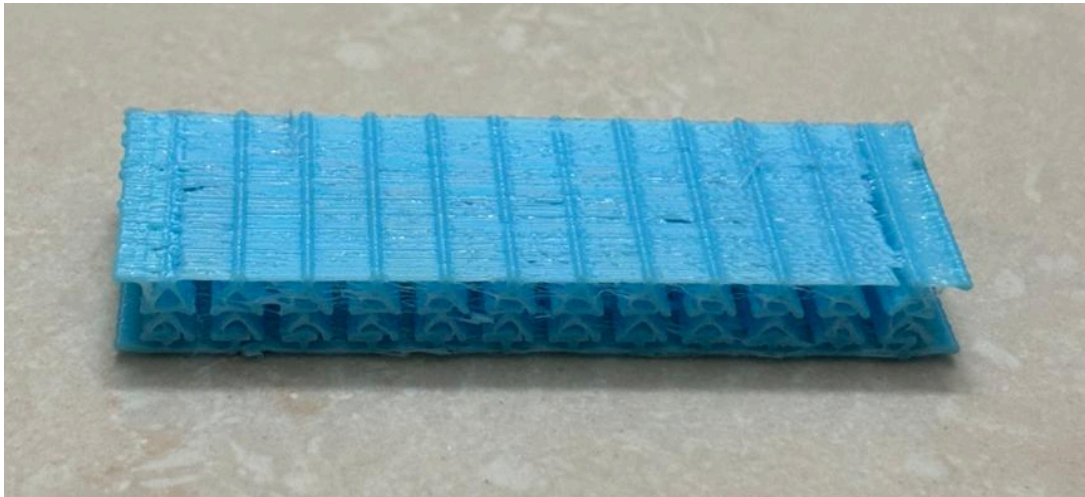


Figure 3.27: DAH test specimen (TPU)



Figure 3.28: Solid test specimen (TPU)

The experimental setup was made of steel and the frame was a vertical guide column. A release mechanism that was controlled manually allowed to control the drop height according to requirements. The impactor weighs 5 kilograms and three different fall heights were taken for each iteration.

An accelerometer positioned under the table on which the specimen was clamped, took the acceleration readings. The model of the accelerometer was 333B30, sensitivity was  $\pm 10\%$ , and frequency range was  $\pm 5\%$ . The sensing element was ceramic. Figure 34 shows the experimental setup that was used.



Figure 3.29: Drop Impact Testing Setup

### 3.3.2 Experimental Procedure

The specimen was positioned on a 2mm rubber pad on the cylindrical table top. Once the specimen was properly clamped, the cylindrical impactor weight was positioned above the specimen, ensuring that the impactor weight covered the whole slice of the protective padding. Afterwards, the cylindrical impactor was dropped on the specimen from three different heights every iteration. The voltage readings were taken from the accelerometer and the force was calculated to compare the performances of the three specimens.

### ABS Solid Block Testing:

Primarily, two solid blocks made of ABS and TPU were tested on experimental setup in order to present the performance of TPU over ABS material. While TPU is generally the better choice for vibration dampening and impact absorption, a comparative test between the two material blocks was conducted to further authenticate our choice of material for the impact testing and possible uses in sports and industrial safety gears.



Figure 3.30: ABS Solid block test setup

Figure 35 shows the test setup for the ABS solid block. Since, all three of the specimens were very small in thickness, the close contact between the metal impactor and the metal anvil or table top would have the possibility of undergoing abrasion and damage. In order to prevent that, a standard sized circular rubber pad with 2 mm thickness was placed beneath the specimens and above the anvil. The specimen was placed in the center of the cylindrical table top and the weight was dropped from three different heights 0.1, 0.25 and 0.5 m respectively.



Figure 3.31: Post-impact ABS solid block

Figure 36 shows the conditions of the ABS block after the post-impact test. The ABS block surface underwent severe cracking and abrasion after just three iterations. A line of crack on the right side of the specimen and right in the middle of its front face was observed after three consecutive impactor drops. The edges of the block were damaged and the solid block showed plastic deformation. This makes it unsuitable for applications in impact absorption or stress distribution. The next two tests were conducted on the solid TPU block and the double arrowhead block and their impact response was compared.

### **TPU Solid Block Testing:**



Figure 3.32: TPU solid block test setup

Figure 37 shows the setup for the solid TPU block under the same drop impact conditions. The cylindrical table top has a 2 mm rubber pad positioned on top of it. The specimen was placed on this rubber pad in order to prevent any kind of surface damage or abrasion. The impactor was positioned above the specimen, ensuring that the drop would provide a uniform strike on the specimen. The weight was dropped from three different heights: 0.1, 0.25 and 0.5 m respectively.



Figure 3.33: Post impact solid TPU block

Figure 38 showcases the condition of the TPU block after multiple drop impact tests in the setting. The surface was checked thoroughly afterwards for residual deformation. The post-impact TPU block showed minor compression marks on it. However, it did not crack or face severe deformation like the ABS block previously. The toughness and elasticity of TPU helped it to recover and reverse the deformation after every impact. This result validates the TPU's ability to absorb and dissipate impact energy sufficiently. The center of the specimen showed comparatively more visible stress marks at the contact region of the impact. In the solid block, due to the absence of special geometric stress dispersion mechanisms, this region undergoes localized deformation. So, despite being of good resilience, the solid TPU block can transmit force to the bottom surface through concentrated stress points.



Figure 3.34: Double Arrowhead TPU specimen test setup

Figure 39 is the experimental setup of the double arrowhead TPU structure under the same impact conditions. For accurate comparative evaluation, similar boundary conditions were maintained in this drop impact test. The cylindrical impactor was at the same height and the DAH specimen was placed on the same rubber padded tabletop. Any abrasion was avoided and an uniform strike by the impactor was ensured. The weight was dropped from three different heights: 0.1, 0.25 and 0.5 m respectively like the previous tests.



Figure 3.35: Post impact double arrowhead TPU

Figure 40 shows the state of the post-impact double arrowhead TPU padding. The structure shows unharmed arrowhead units. The surface has minimal visible deformation as the stress was distributed through the interconnect arrowhead joints. It reduced the force transmitted to the bottom layer and made the structure recover entirely after each drop impact test. The arrowhead unit cells at the impact region showcased temporary lateral expansion and compression but no permanent cracks or warping were present. However, delamination is observed in merging zones in the upper plate, possibly due to low-resolution print quality and less layer adhesion while printing in the upper plate.

## Chapter 4: Results and Discussions

### 4.1 Analysis of the study

The study shows how protective paddings with a Double Arrowhead auxetic structure perform under impact conditions. The simulation results included spreading out stress, and improved stress distribution in this auxetic structure. To testify these results, 3D-printed TPU samples were tested in the lab. The behaviour of the TPU specimens performed better, supporting the accuracy of the models. The findings also showed that this design offers a good mix of being lightweight, comfortable, and protective, making it a strong option for future sports and industrial gear.

#### 4.1.1 Experimental Results

The accelerometer was calibrated such that 100 mV voltage corresponded to 9.8 m/s<sup>2</sup>.

##### ABS solid block + Reference Rubber pad test results:

| Drop Height | Acceleration (a) in inner surface (m/s <sup>2</sup> ) | Force transmitted to inner surface (kN) |
|-------------|-------------------------------------------------------|-----------------------------------------|
| 0.1 m       | 68.67                                                 | 1.71675                                 |
|             | 73.575                                                | 1.839375                                |
|             | 78.48                                                 | 1.962                                   |
| 0.25m       | 274.68                                                | 6.867                                   |
|             | 294.3                                                 | 7.3575                                  |
|             | 353.16                                                | 8.829                                   |
| 0.5         | 412.02                                                | 10.3005                                 |
|             | 412.02                                                | 10.3005                                 |
|             | 412.02                                                | 10.3005                                 |

Table 6: ABS solid block test results

The ABS solid specimen started showing signs of damage and the destruction of the layers as early as after dropping the impactor on it from a height of 0.1m. The structure failed abruptly

by cracking after the impactor dropped on it from the height of 0.25m. The data collected for the drop heights 0.25 and 0.5m are not usable as the test would require an unharmed specimen for correct data. Since ABS is a very rigid structure, it introduced minimum friction with the anvil. The force transmitted to the inner surface however was very high.

**TPU solid block + Reference pad test results:**

| Drop height | Acceleration in inner surface (m/s <sup>2</sup> ) | Force transmitted to inner surface (kN) |
|-------------|---------------------------------------------------|-----------------------------------------|
| 0.1m        | 37.278                                            | 0.93195                                 |
|             | 37.278                                            | 0.93195                                 |
|             | 38.259                                            | 0.956475                                |
| 0.25m       | 127.53                                            | 3.18825                                 |
|             | 137.34                                            | 3.4335                                  |
|             | 137.34                                            | 3.4335                                  |
| 0.5m        | 0                                                 | 0                                       |
|             | 0                                                 | 0                                       |
|             | 0                                                 | 0                                       |

Table 7: TPU solid block test results

The TPU solid block showed better performance in comparison to the ABS solid block. The transmitted force to the inner plate is much lower compared to the ABS solid block, The results supported the choice of TPU as a fitting choice for protective paddings.

**TPU double arrowhead padding + Reference pad test results:**

| Drop height | Acceleration in inner surface (m/s <sup>2</sup> ) | Force transmitted to inner surface (kN) |
|-------------|---------------------------------------------------|-----------------------------------------|
| 0.1m        | 28.449                                            | 0.711225                                |
|             | 37.278                                            | 0.93195                                 |
|             | 44.145                                            | 1.103625                                |

|       |        |        |
|-------|--------|--------|
| 0.25m | 117.72 | 2.943  |
|       | 137.34 | 3.4335 |
|       | 156.96 | 3.924  |
| 0.5m  | 0      | 0      |
|       | 0      | 0      |
|       | 0      | 0      |

Table 8: TPU double arrowhead padding test results

In case of the first two height drops, the TPU double arrowhead structure maintained an intact condition. However, the available 3D printing technology produced poor lamination on the top plates at the merging zones with the unit cells. The first two iterations displayed significantly lower force transmission to the inner surface compared to both the solid ABS and TPU blocks. The third impact showed higher force transmission due to the top plate delamination. Critical to highlight, The double arrowhead unit cells remained unharmed after the repeated impacts and this proves the structures ability to efficiently absorb impact. High resolution printing output and better layer adhesion will ensure even better performance and outcomes.

## 4.2 Application of this study in the field of Mechanical and Industrial engineering practice

From an academic perspective, this research makes a substantial contribution to mechanical and industrial engineering practice by integrating state-of-the-art material science with computational modeling and additive manufacturing techniques. The precise application of the Double Arrowhead auxetic geometry provides a robust framework for mechanical engineers, demonstrating how material structure can be discreetly manipulated to achieve simultaneous improvements in both the strength-to-weight ratio and the energy endurance efficiency of protective systems.

Through a controlled simulation and experimental validation, the study shows the way a structural geometry can be engineered to restrain deformation and stress dissipation under impact, which is fundamental in mechanical design. Moreover, the integration of Fused

Deposition Modeling (FDM) makes manufacturing of the customized, lightweight components with complex geometries easy, which cannot be easily achieved with traditional methods. The outcomes can guide engineers in designing more secured and more efficient products in sports equipment, automotive interiors and exteriors, industrial safety gear and even aerospace applications where impact resistance, material efficiency and ergonomic design are critical.

# Chapter 5: Conclusion

## 5.1 Conclusion

This study successfully proved that auxetic structures, in particular, the double arrowhead (DAH) geometry, exhibit superior impact energy absorption due to their negative poisson's ratio (NPR). DAH structure absorbed and distributed stresses more evenly under compressive or impact loading conditions and minimized localized stress concentrations. Compared to conventional structures like solid TPU or ABS which show a positive poisson's ratio, DAH geometry provides superior impact mitigation, resilience and recoverability.

This study successfully validated the impact response of a 4 X 4 array of double arrowhead structure from a recently published study. The observed results, where the percentage of error was acceptably low, supported the claims of previous studies. The experimental impact testing followed by the Finite Element Analysis validated the performance of a double arrowhead protective padding in comparison to the solid TPU padding. Two solid (TPU and ABS) and one DAH-patterned TPU specimens were fabricated with the help of Fused Deposition Modeling (FDM) and experimented using a cylindrical drop impactor of 5 kg.

The simulation results were identical to the experimental results of the models. The stress on the bottom plate of the model for solid TPU decreased from 0.05 MPa to 0.02 MPa to the DAH model structure which provided 60% better stress distribution. Even though the DAH experienced slightly higher deformation than solid TPU, it was reversible and the residue deformation was negligible. From the experimental results, the ABS bottom plate acquired a transmitted force of 1.72–1.96 kN along with permanent cracks and surface indentation. Solid TPU transmitted only 0.93–0.96 kN of force to the inner surface which reduced the impact force 46%–51% from the ABS specimen. However, the DAH structure shows a significantly improved result with a transmitted force of only 0.71–1.10 kN, which is a 14%–24% improvement from the solid TPU and almost 60% better than the ABS specimen.

These results solidify the double arrowhead structure's ability to mitigate impact force, minimize localized stress, enhance resilience and improve recoverability under repeated load impacts which strengthens the usability of DAH structure in industrial and sports protective gear or clothing. Protective equipment padding such as knee or shin guard paddings with a

solid tpu structure inside weighs 11.07 grams. Integrating DAH structure inside will make the padding 67.4% lighter with a weight reduction of 7.47 grams. This lightweight structure is a critical factor for comfort in wearable safety sports or industrial gear.

## **5.2 Recommendation for future works**

While this study used single-material double arrowhead structures, future research can concentrate on hybrid or composite materials to produce double arrowhead structures. If fiber-reinforced thermoplastics are combined, such as TPU–EVA, or TPU–silicone, better results can be obtained in terms of energy absorption, mechanical performance and comfort. A few other possible research paths are experimenting with nanocomposite modifications using carbon nanotubes, graphene or silica nanoparticles. Hybridization of different materials can improve the performance of the double arrowhead. The hybridization of double arrowhead structures with other auxetic structures may also open doors for interesting studies.

The key area to work on in the future is integrating this double arrowhead structure in protective sports and industrial gear for better energy absorption. Developing knee, elbow, shin guards, body armor and chest protectors, helmet liners with the double arrowhead inside as the inner cushioning layer should be the main focus. Moreover, several industrial safety gears such as protective gloves and sleeves, safety helmets, bump caps, and vibration damping panels can be produced with this structure inside to ensure better energy dissipation. To perform real-time monitoring of the performance, sensor-integrated smart systems can be integrated in commercialized versions of these protective gears. Integrating piezoresistive elements in the padding with the double arrowhead structure can track the impact force and fatigue during sports activities. Establishing active warning systems to alert of excessive impact on body parts and automated replacement alerts to inform of exceeded impact thresholds can improve the product efficiency.

Even though present finite element models provide insightful information, advanced finite element and multiscale modeling can showcase better simulations. Integrating rate-dependent plasticity, nonlinear viscoelasticity, and damage evolution can simulate real impact events more accurately. No hyperelastic behaviour of the double arrowhead structure was studied. This opens many doors and opportunities for further research in related fields.

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